USER MANUAL · WORKED EXAMPLE

Room fire, from empty model to results

One complete simulation in the PyFDS desktop application, with a screenshot of every step: set up the project, mesh the domain, draw the room, define the fire, place instruments, check the model, run the real FDS solver, then explore the results in 3D: slices, flame, isosurfaces, plots, screenshots and an MP4 animation.

ABOUT THIS CASE

A gas burner is lit in the middle of a small room, and we watch what happens for the first 20 seconds. The room is 4.8 × 3.3 m and 2.4 m high, with plasterboard walls and ceiling, a door in the south wall and a window in the east wall. The burner grows from nothing to about 360 kW over 15 seconds. A heat detector under the ceiling above the fire and four thermocouples on a vertical line record the room's response.

The case answers three practical questions: how fast the fire grows, how soon a ceiling heat detector would respond, and how hot the gas gets at different heights. It is small and quick (about three minutes to compute) so every step can be shown end to end.

Application: PyFDS desktop Solver: FDS 6.10.1 Viewer: Smokeview 6 Case: demo_room
Run date: 11 Oct 2026

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The case at a glance

370 kW
peak heat release rate

≈ 10 s
heat detector activates (FDS reports 9.7 s)

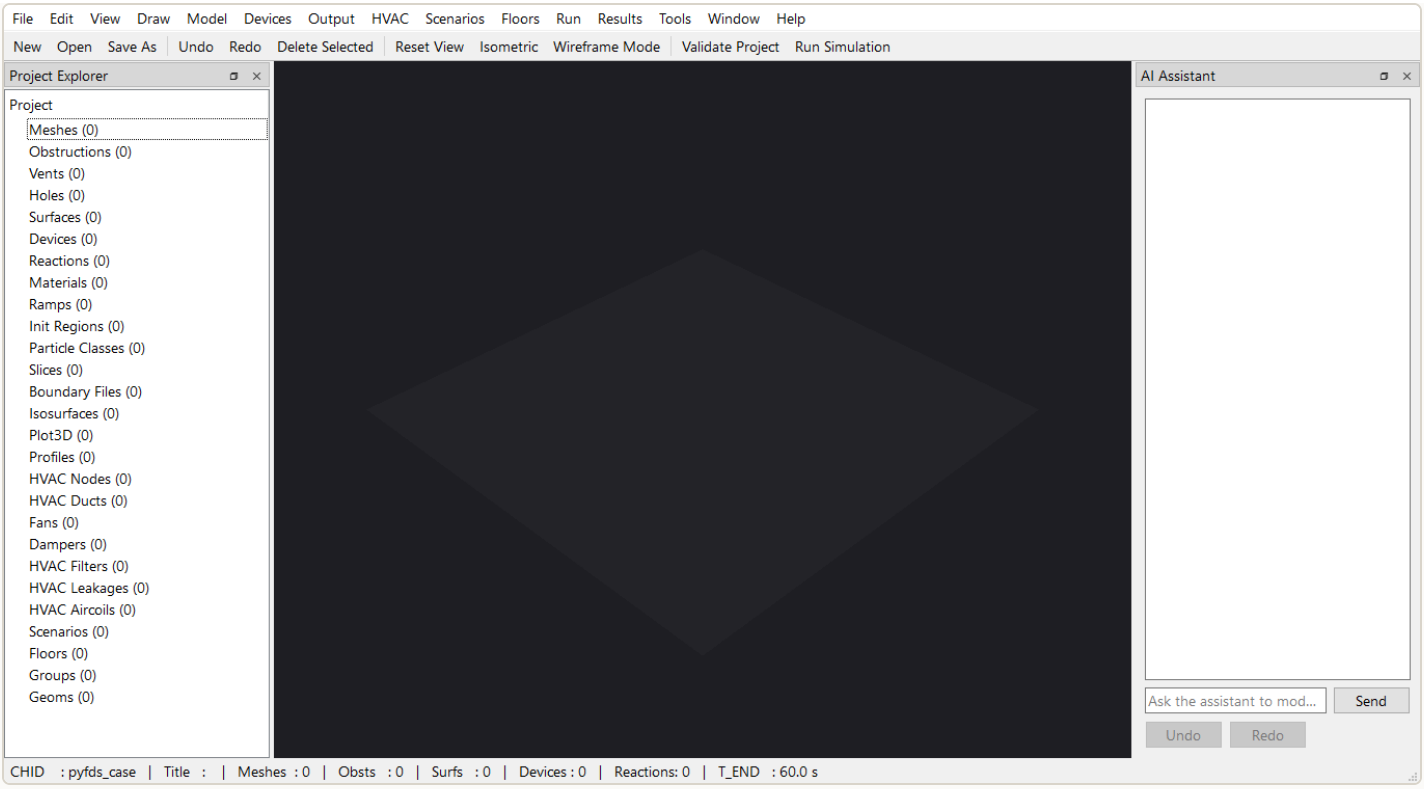
707 °C
gas temperature 0.6 m above the floor at 20 s

160 s
time FDS took to run the 20 s of fire

ITEM	VALUE
Room	4.8 × 3.3 m floor, 2.4 m high; 0.15 m double-layer gypsum walls and ceiling
Openings	Door 0.9 × 2.1 m (south wall); window 1.2 × 1.05 m, 0.9 m above the floor (east wall)
Fire	Propane burner 0.6 × 0.6 m, 0.15 m high, 500 kW/m² on every exposed face (0.72 m², about 360 kW at full size), growing with t² to full size at 15 s
Calculation domain	6.0 × 4.5 × 3.0 m, open on all sides and above; 40 × 30 × 20 cells of 0.15 m (24,000 cells)
Instruments	Heat detector (74 °C, RTI 5) under the ceiling; thermocouples at 0.6, 1.2, 1.8 and 2.3 m
Output pictures	Temperature slices through the fire, every second
Simulated time	20 s

The application window

Start PyFDS with `python main.py`. A new, empty project opens. Everything in this manual is done from this window.



The empty project. **Menus** across the top (File, Edit, View, Draw, Model, Devices, Output, Run, Results ...). **Toolbar** with the common actions, including Validate Project and Run Simulation. **Project Explorer** on the left lists every object in the model by type. The **3D view** is in the middle. The **status bar** along the bottom shows the case name and object counts. The AI Assistant panel on the right is not used in this walkthrough.

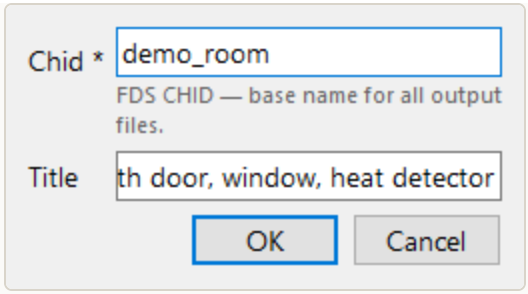
Every object you add appears in the Project Explorer and the 3D view at once. Every action can be undone with Edit > Undo or the toolbar button.

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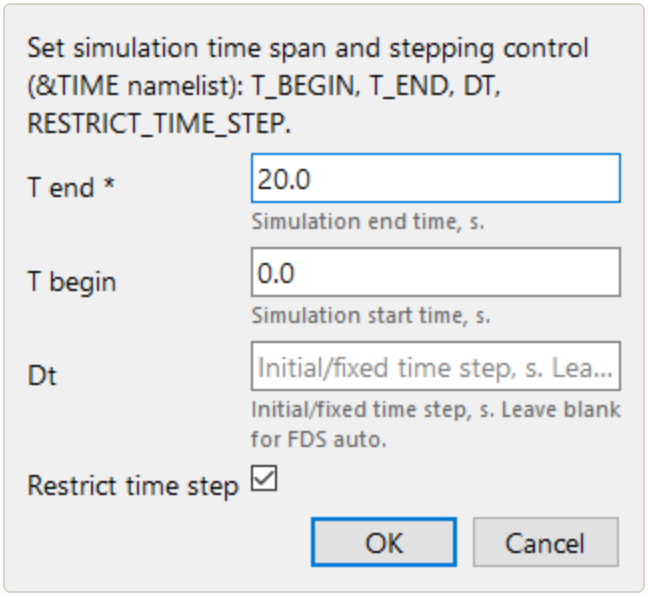
1 · Project and simulation setup

`MENU` File > Project Settings..., then Model > Simulation Parameters...

Give the case a name and set how long to simulate. The **case ID** (CHID) becomes the first part of every output file name, so keep it short and without spaces.



Project Settings: case ID `demo_room` and a title.



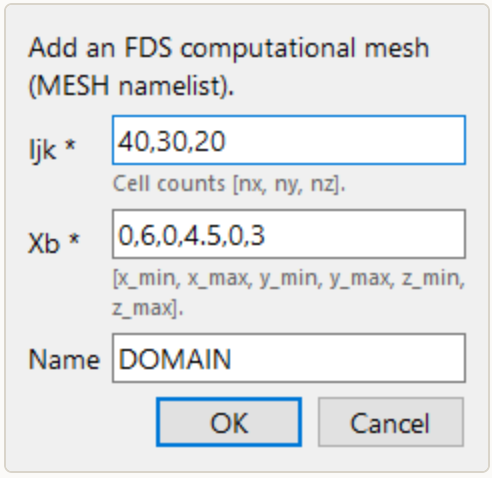
Simulation Parameters: end time 20 s. The other values keep their defaults.

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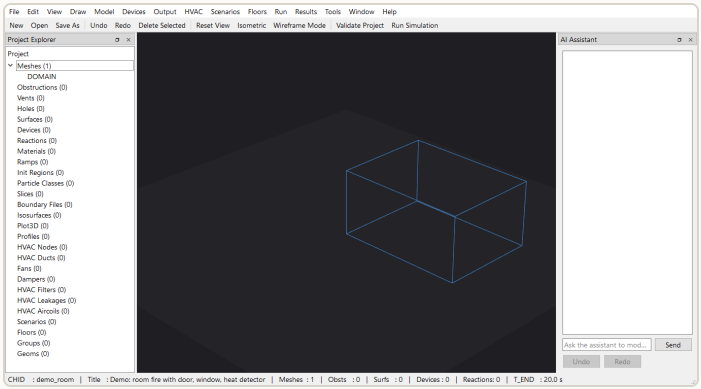
2 · Mesh

MENU Model > Add Mesh...

The mesh is the box FDS calculates in, divided into small cells. Enter the cell counts and the box corners. Here the box is 6.0 × 4.5 × 3.0 m, divided into 40 × 30 × 20 cells, so each cell is 0.15 m. That size puts the room walls exactly on cell boundaries.



Add Mesh. **Ijk** = cells in x, y, z. **Xb** = x min, x max, y min, y max, z min, z max in metres.



The mesh appears as a blue frame in the 3D view.

Cell counts are checked for you

FDS needs each cell count to be made of the factors 2, 3 and 5 (40, 30 and 20 are). The model check in step 6 flags a count such as 14 or 17 before anything is run.

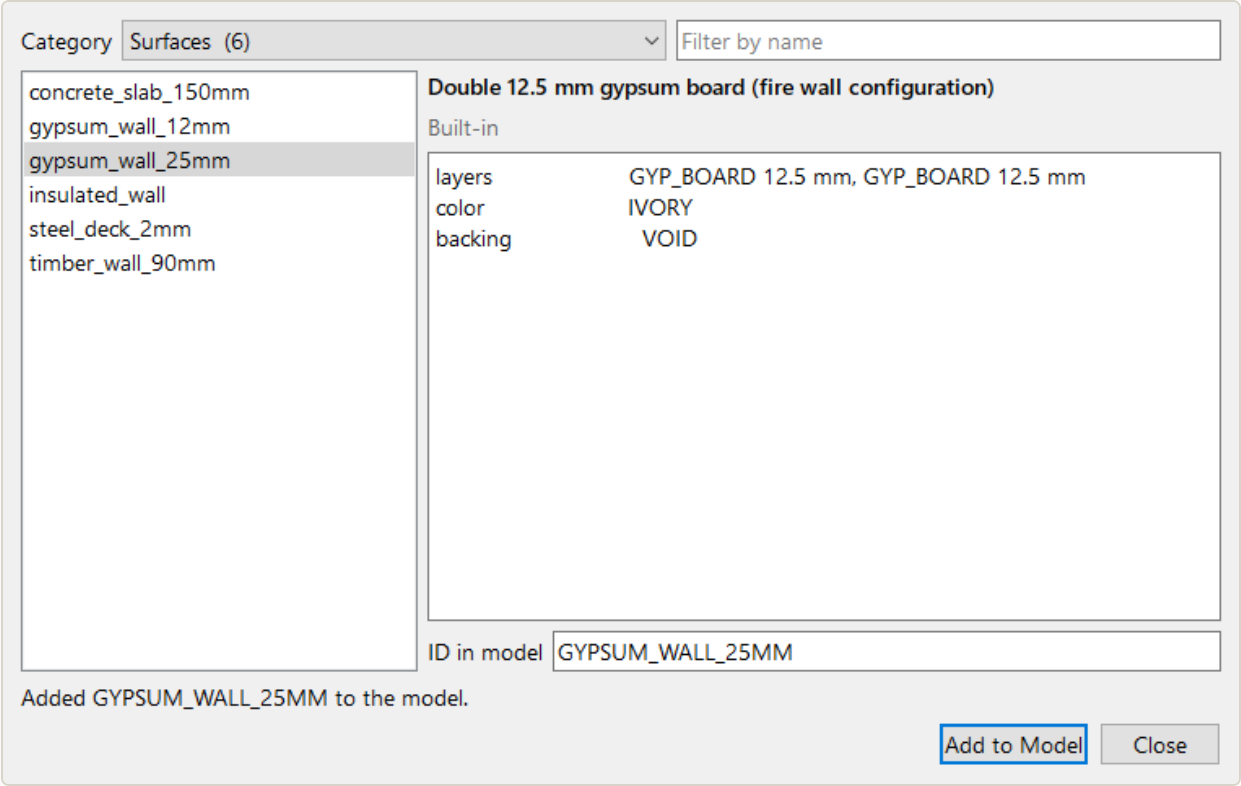
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3 · Geometry

3a · Choose the wall construction from the library

`MENU` Model > Library...

The library holds ready-made materials, wall constructions, fuels, detectors and more. Pick **Surfaces**, select `gypsum_wall_25mm` (two 12.5 mm gypsum boards), and press **Add to Model**. The gypsum board material it is made of comes along automatically.

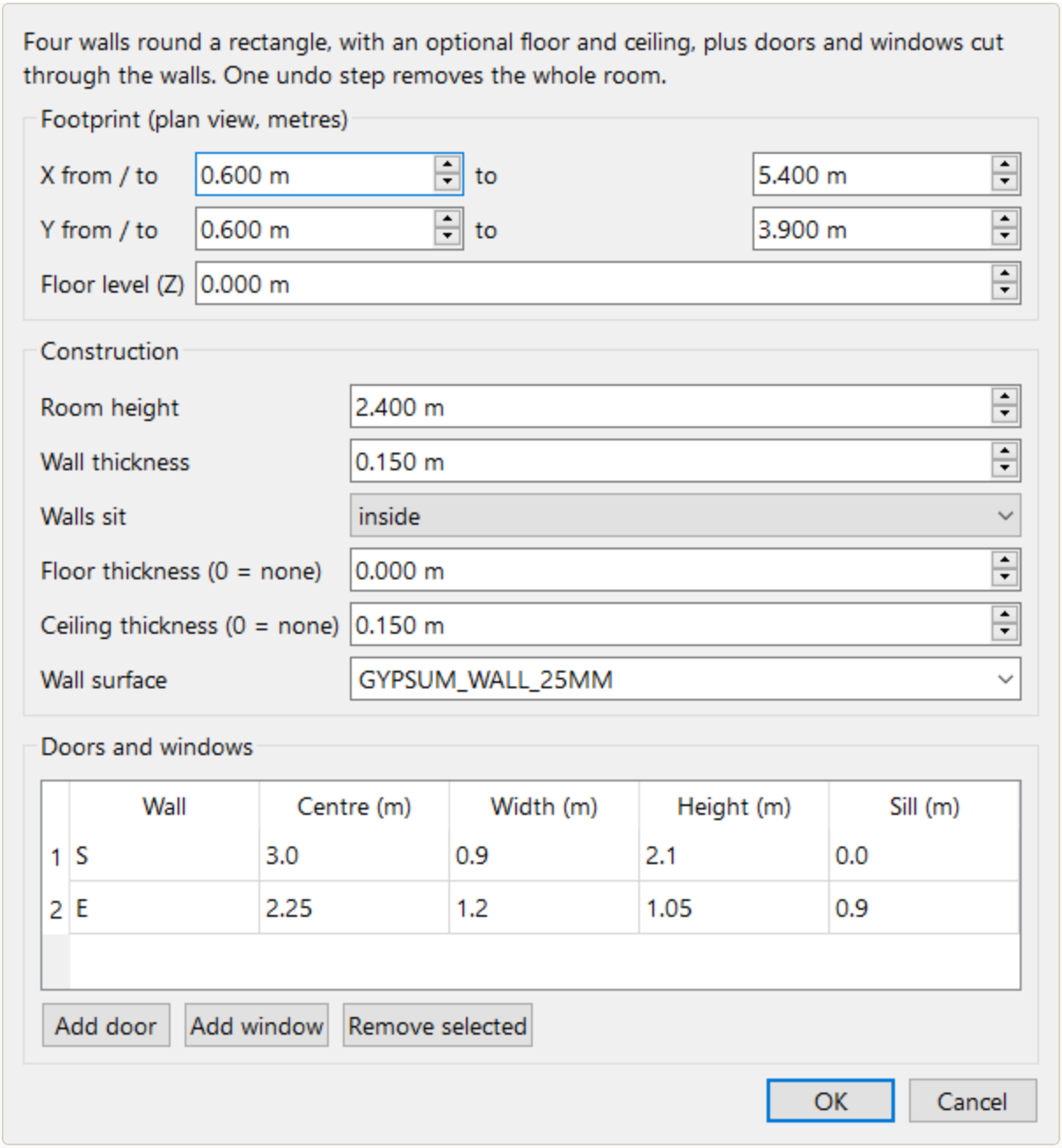


The library browser. The right side shows what the entry contains. “Added GYPSUM_WALL_25MM to the model” confirms it.

3b · Draw the room

`MENU` Draw > Draw Room (walls, door, window)...

Describe the room once: the outline, height, wall thickness and the wall surface. Add the door and window in the table. **Wall** is S, E, N or W; **Centre** is the distance along that wall; **Sill** is the height of the bottom edge above the floor.



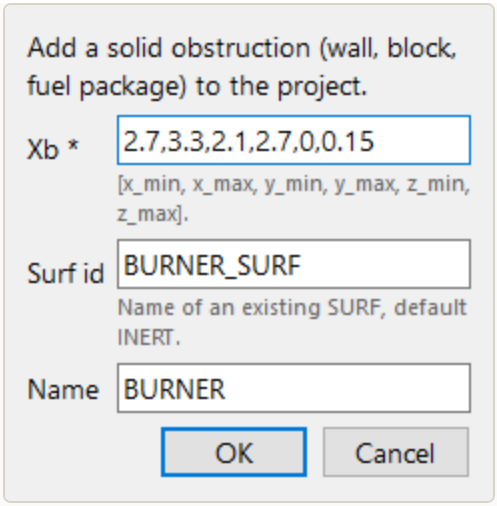
Draw Room. The door is on the south wall, 3.0 m along, 0.9 × 2.1 m. The window is on the east wall, 2.25 m along, 1.2 × 1.05 m, sill at 0.9 m.

One press of OK builds four walls and a ceiling and cuts both openings. The whole room is one undo step.

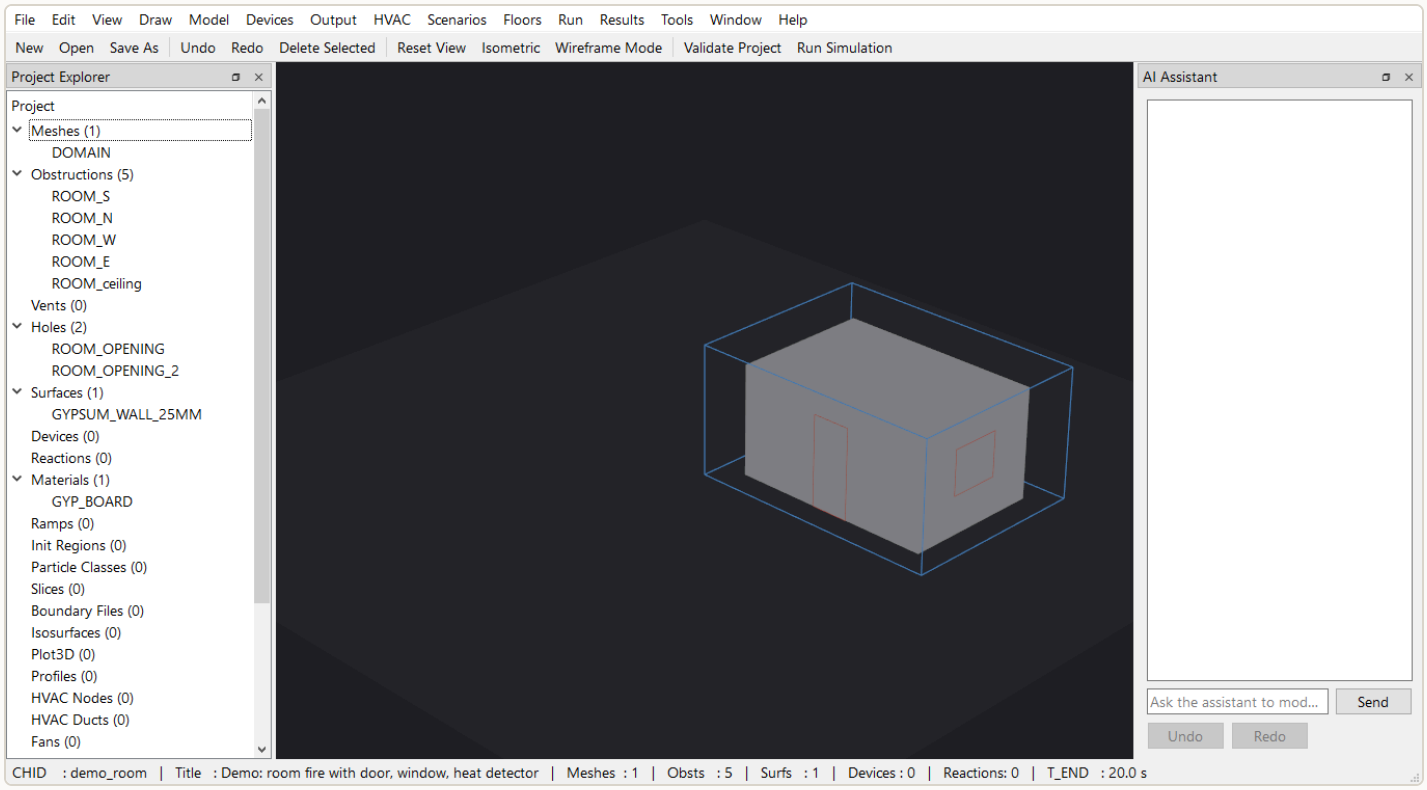
3c · The burner block

`MENU` Model > Add Obstruction...

The burner is a low block on the floor in the middle of the room, 0.6 × 0.6 m and 0.15 m high. It uses a surface called `BURNER_SURF`, which is defined in the next step.



Add Obstruction: position and size in **Xb**, and the surface it uses.



The room in the 3D view: walls and ceiling in grey, door and window outlined in red, mesh frame in blue. The green faces that mark the open boundaries are hidden here (View > Show/Hide > Vents) so the room can be seen.

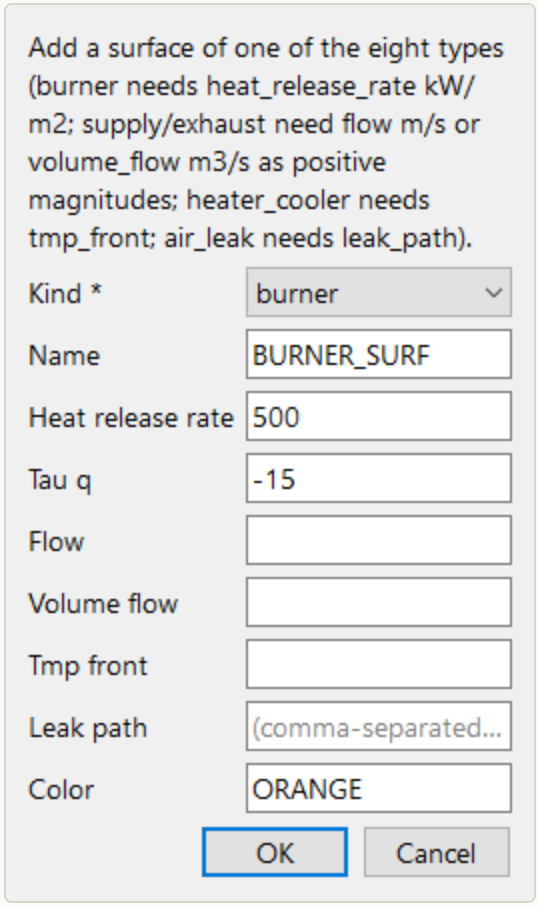
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4 · Physics: fire and boundaries

4a · The burner surface

MENU Model > Add Typed Surface...

Choose the surface type **burner** and give its heat release per square metre (500 kW/m²). A negative **Tau q** of −15 means the heat output grows with the square of time and reaches full strength after 15 seconds. The dialog asks only for what a burner needs.

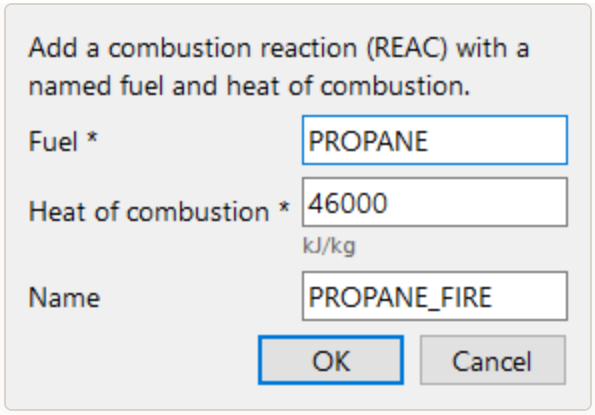


A burner surface: 500 kW/m², t-squared growth over 15 s.

4b · The fuel

MENU Model > Add Reaction...

Say what burns: propane, with its heat of combustion of 46,000 kJ/kg. FDS uses this, together with the burner's heat release, to work out how much gas to feed in and how much smoke it makes.

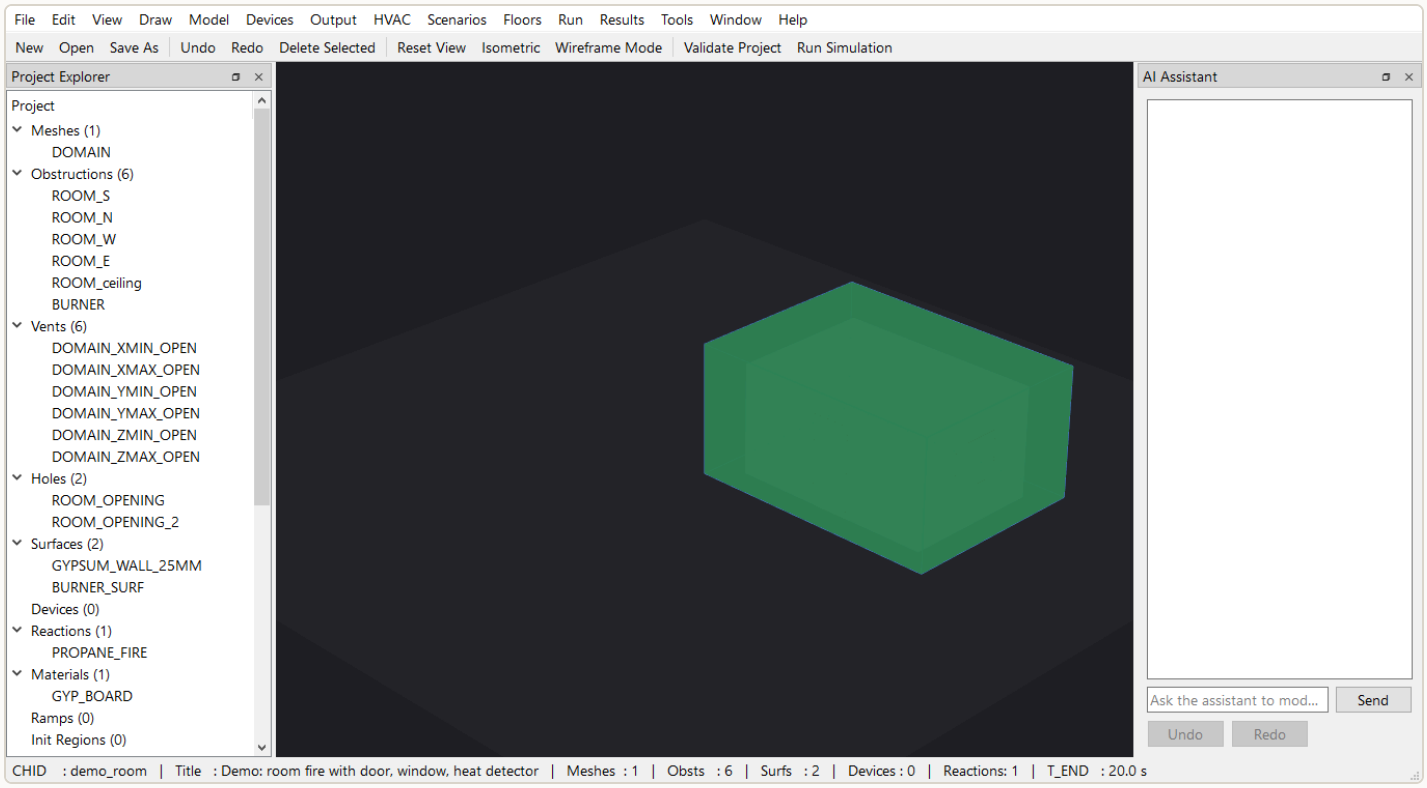


The reaction: fuel PROPANE, heat of combustion 46,000 kJ/kg.

4c · Open the outside

`MENU` Model > Open Mesh Boundaries

One click closes the mesh with open vents on all six faces, so air can come in and hot gas can leave. Six vents appear in the Project Explorer.



Six `DOMAIN_*_OPEN` vents were created, one per face. The 3D view shows them as green translucent faces.

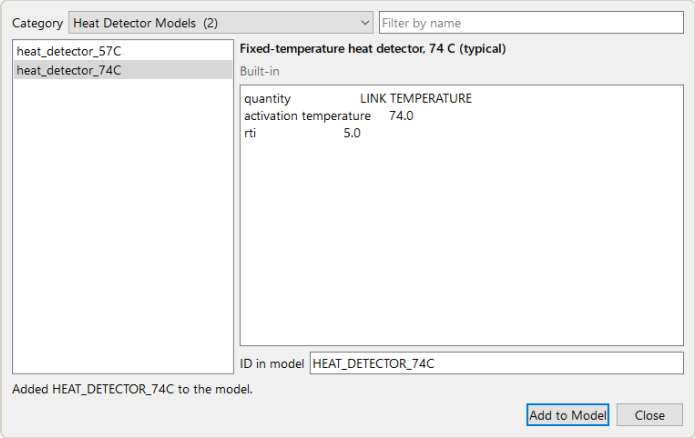
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5 · Instruments and outputs

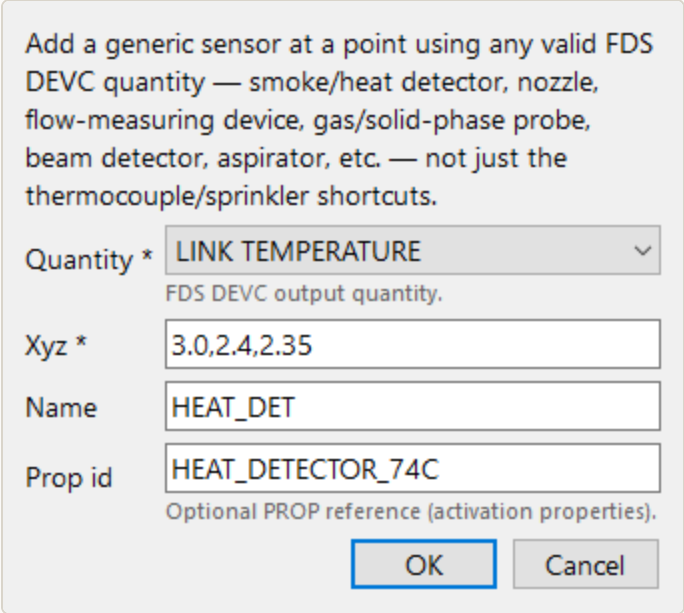
5a · Heat detector

MENU Model > Library... (Heat Detector Models), then Devices > Add Device (by Quantity)...

First add the detector's properties from the library: a heat detector that activates at 74 °C with a response time index (RTI) of 5. Then place a device that uses it, under the ceiling above the fire.



Library, Heat Detector Models: `heat_detector_74C`.



The detector device at x 3.0, y 2.4, z 2.35 m, using the property just added.

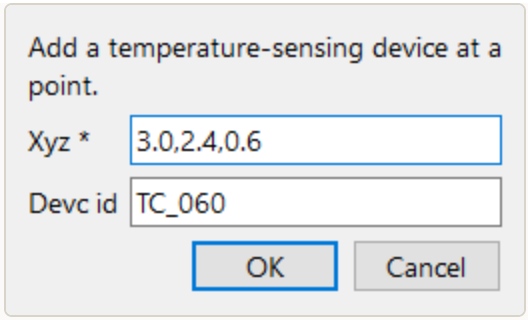
Check library values for your own work

The detector entry is a typical value, supplied as a starting point. Replace it with the data of the product you are modelling.

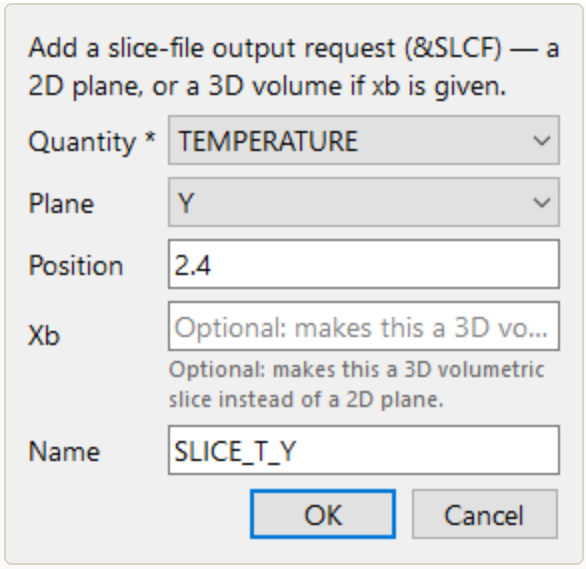
5b · Thermocouples and slices

MENU Devices > Add Thermocouple..., and Output > Add Slice...

Four thermocouples on a vertical line above the fire (0.6, 1.2, 1.8 and 2.3 m) record the gas temperature over time. A **slice** is a flat picture plane that FDS saves every second. Two temperature slices were added, one on the plane y = 2.4 m and one on x = 3.0 m, both through the fire.



Thermocouple at x 3.0, y 2.4, z 0.6 m. Repeat for 1.2, 1.8 and 2.3 m.



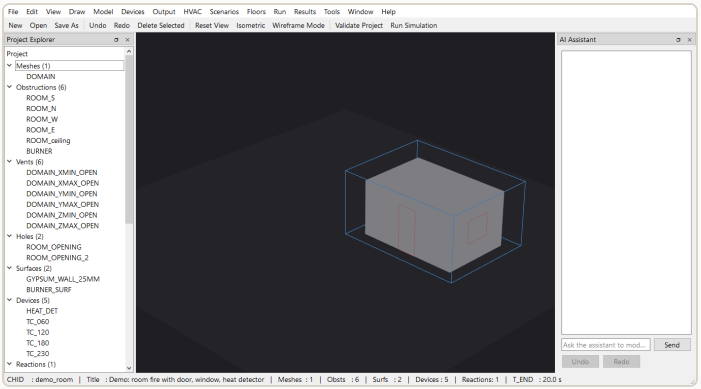
Temperature slice on the plane y = 2.4 m.

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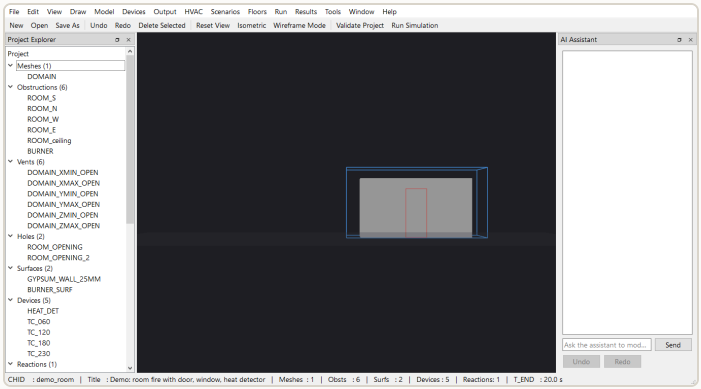
6 · Check the model

MENU Model > Validate Project (also on the toolbar)

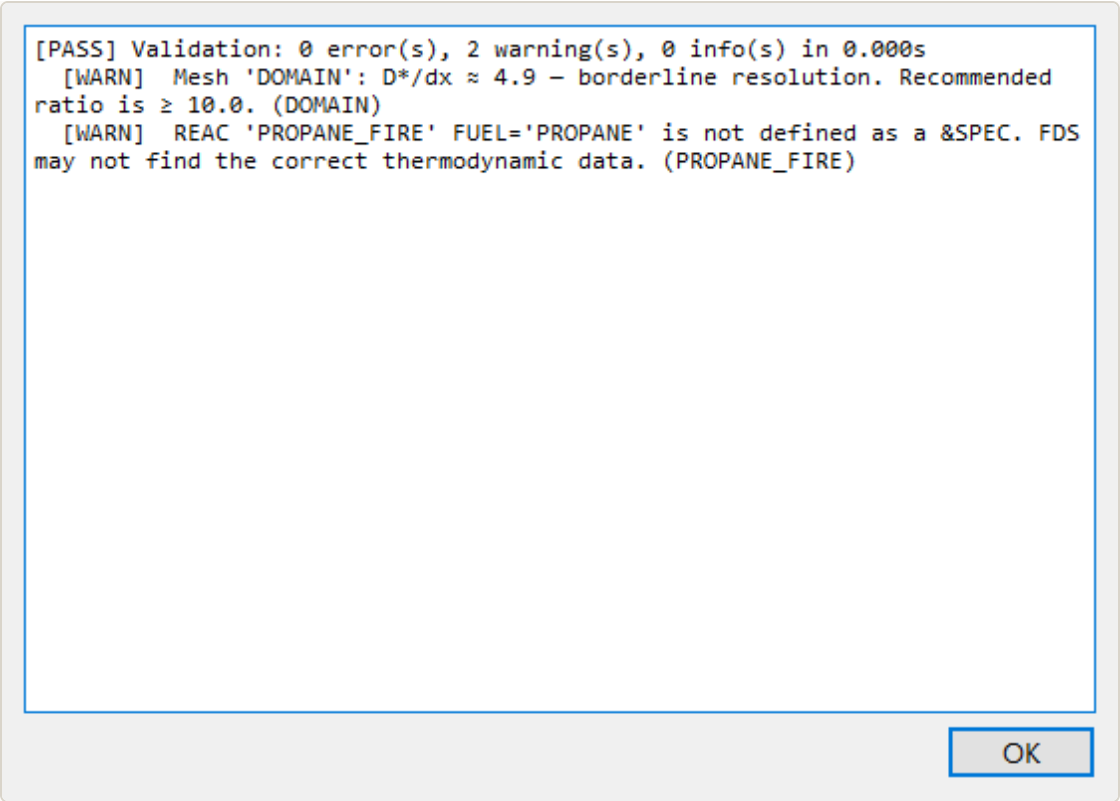
Before running, look at the finished model from a few angles (View > Standard Views), then run the model checks. PyFDS applies 28 checks and reports errors, which stop a run, and warnings, which do not.



Isometric view of the finished model. The Project Explorer shows every object.



Front view: the door is visible on the south wall.



The validation report: **0 errors, 2 warnings**, so the run can go ahead.

Warning 1 · Fire resolution is borderline

The fire-resolution ratio (D*/dx) is 4.9; 10 or more is recommended for results you rely on. Halving the cell size would reach that but takes about eight times as long to compute. Acceptable for a demonstration.

Warning 2 · Fuel species not defined

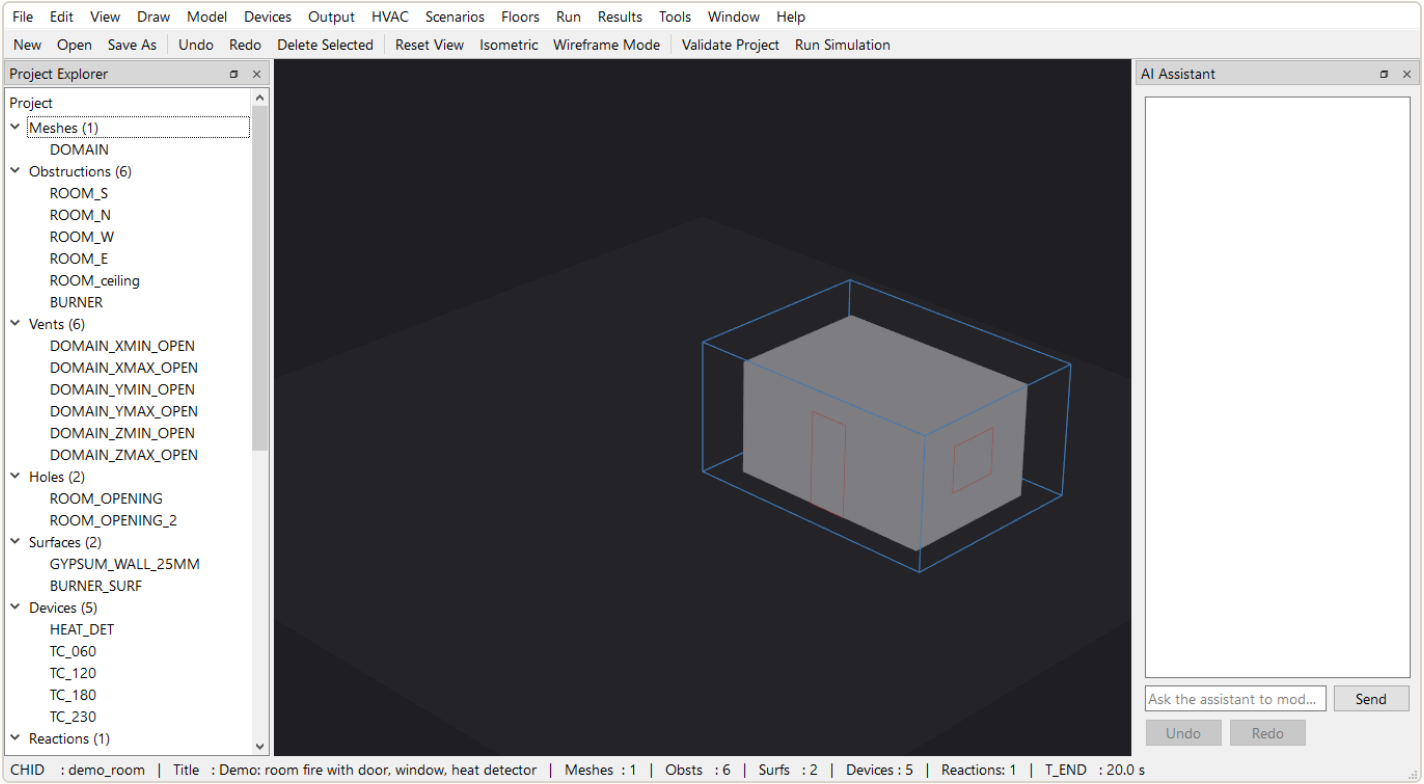
PROPANE is named without its own species entry. FDS knows propane built in, so this run is unaffected.

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7 · Save

`MENU` File > Save As... (or the toolbar)

Save the project as a `.pyfds` file. It holds everything: geometry, library items, settings. Open it again later with File > Open.



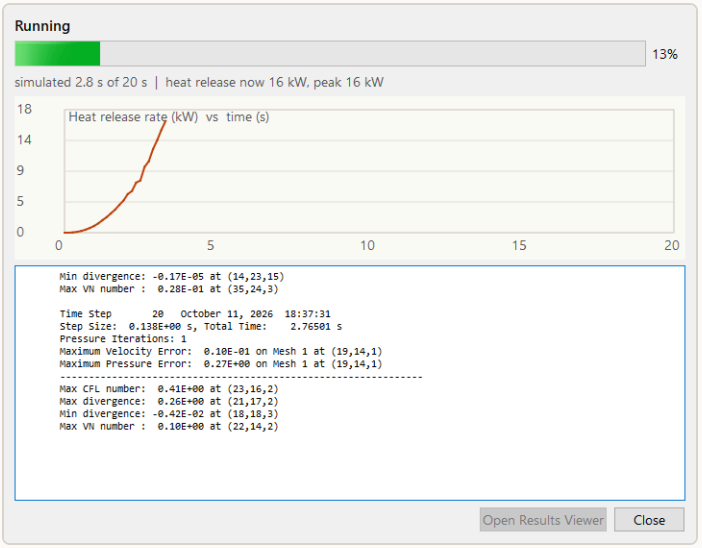
Saved as `demo_room_gui.pyfds`.

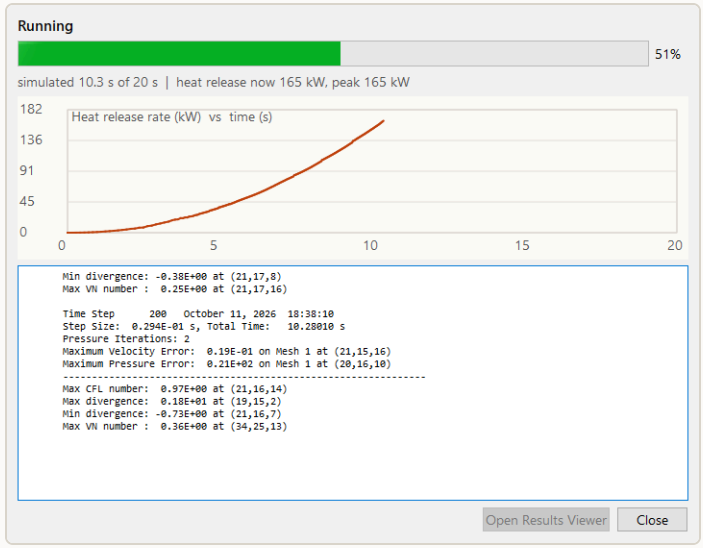
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8 · Run

MENU Run > Run Simulation... (or the toolbar). Choose an output folder and the number of processes (1 here).

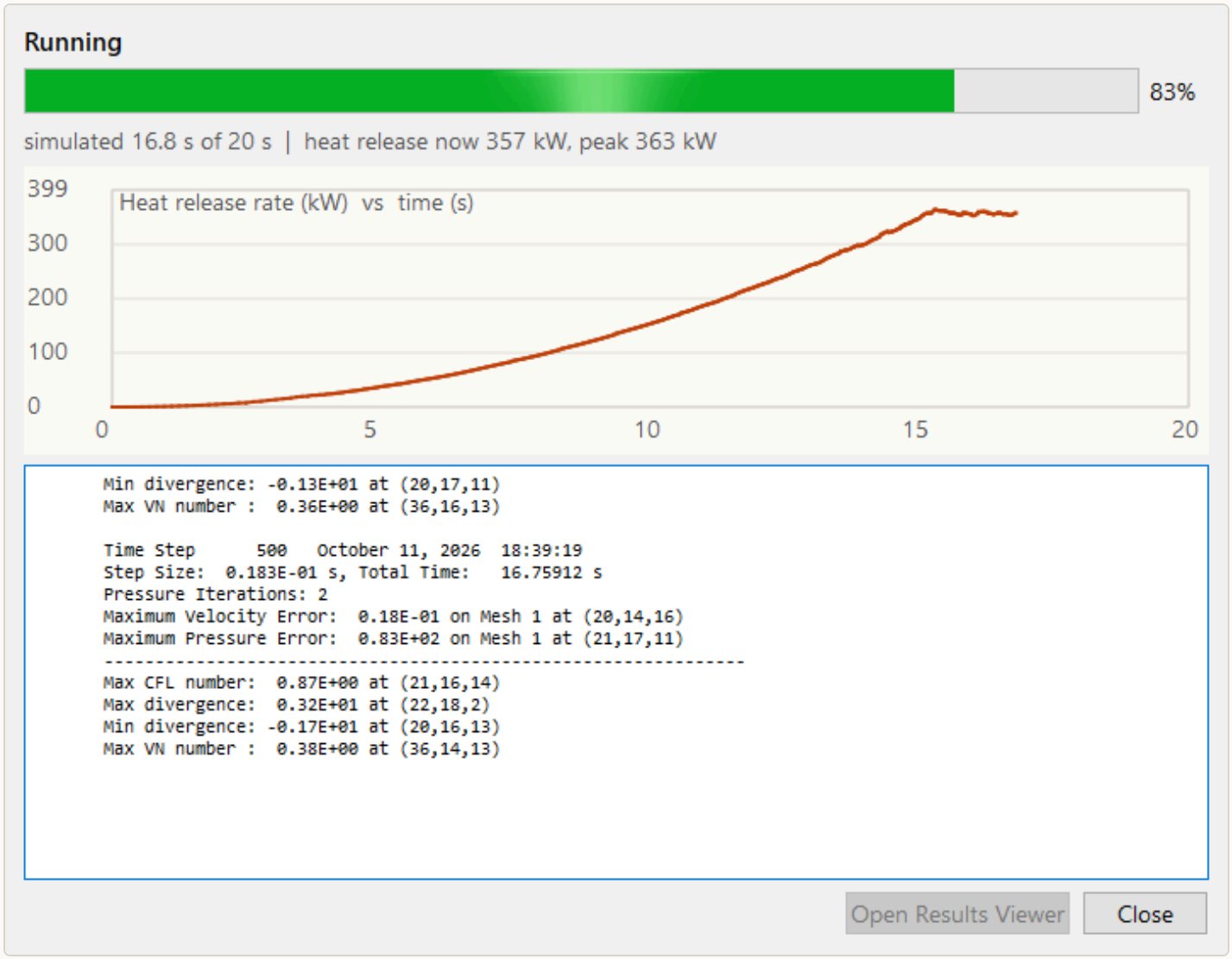
PyFDS checks the model again, writes the FDS input file, and starts the real FDS solver. The **Simulation Monitor** opens by itself. It shows progress, the heat release rate as it develops, and the solver's own log. You can keep working in the main window or close the monitor; the run continues.



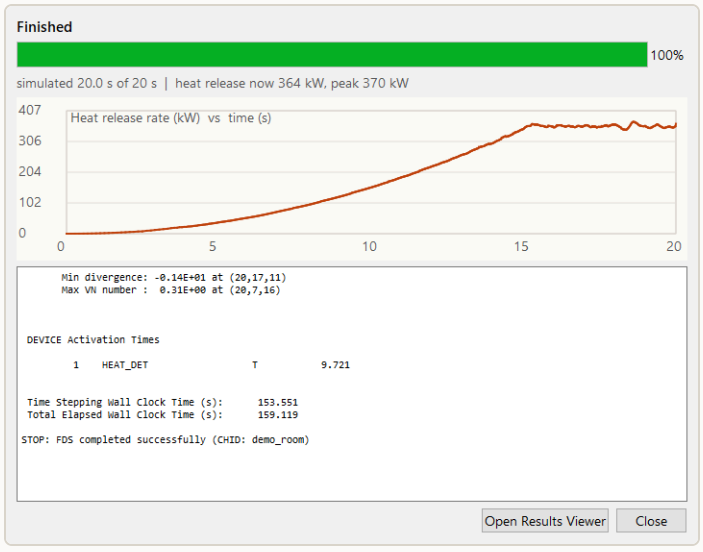


Early in the run.

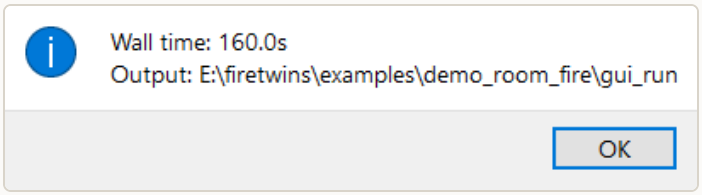
Fire growing along the t-squared curve.



Close to full size: the heat release rate levels off near 360 kW.



Finished. The log shows the heat detector's activation time and “FDS completed successfully”. **Open Results**



A message confirms completion. This run took 160 s of computer time.

Viewer is now available.

If FDS rejects the input, the run is reported as failed together with FDS's own error message, and the monitor shows “FDS stopped with an error”.

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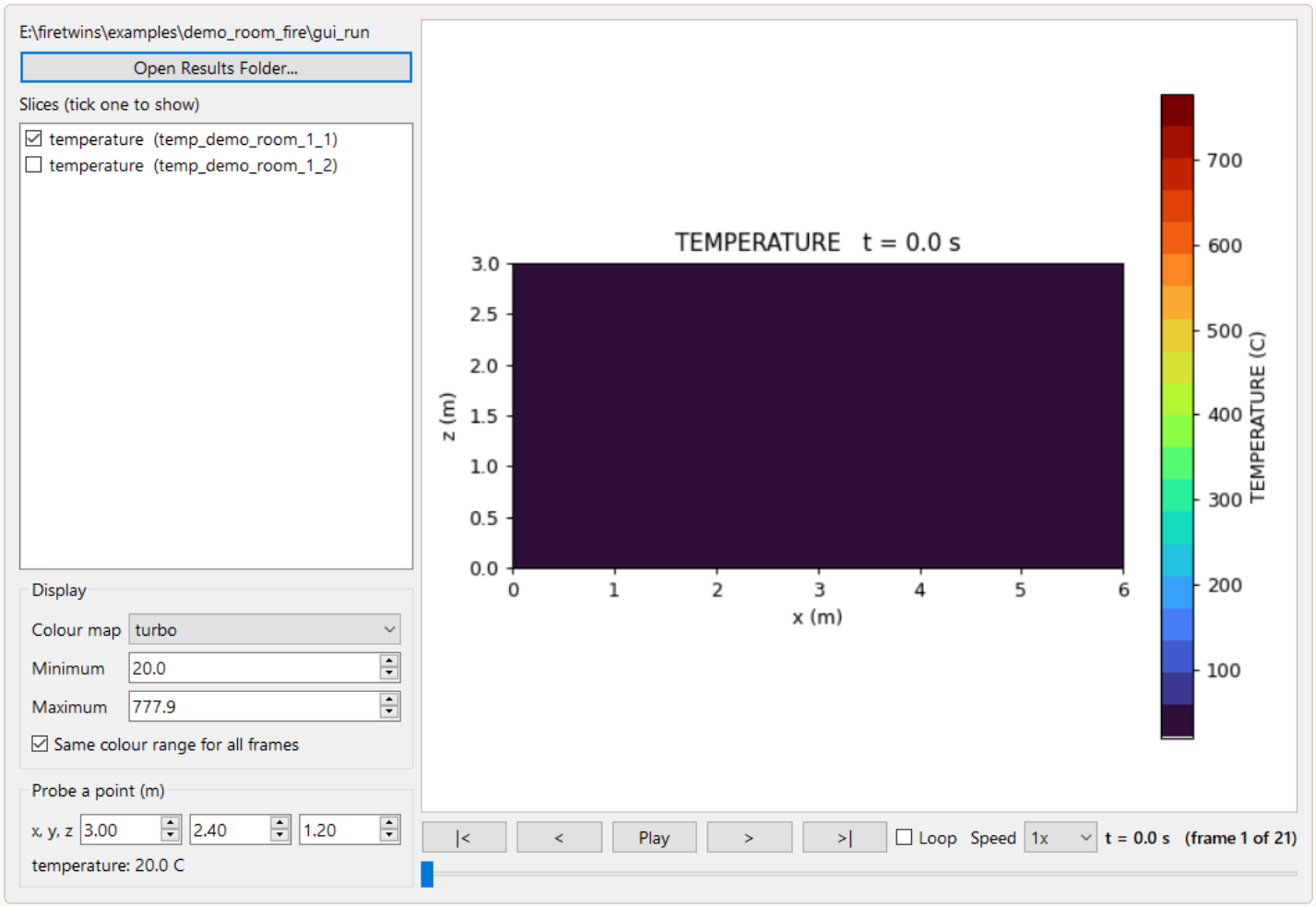
9 · Results: a first look

This section uses the simple 2D results window. The full 3D viewer, with smoke, isosurfaces, animation and more, is in section 10.

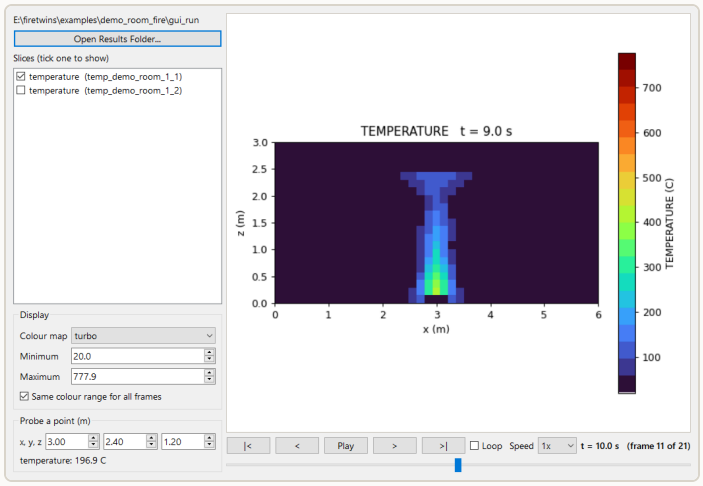
9a · Quick results viewer: play the run back

`MENU` Results > Results Viewer... (or the button in the monitor)

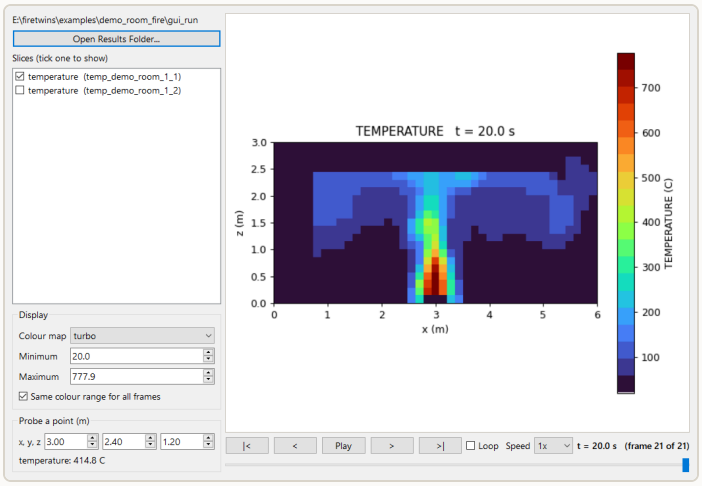
The viewer plays the saved slices back like a video. Use the controls at the bottom: jump to the start or end, step one frame, **Play** and **Pause**, speed from 0.25× to 10×, loop. Drag the slider to any time. The colour map and range can be changed on the left, and **Probe a point** reads the temperature at any position on the plane shown.



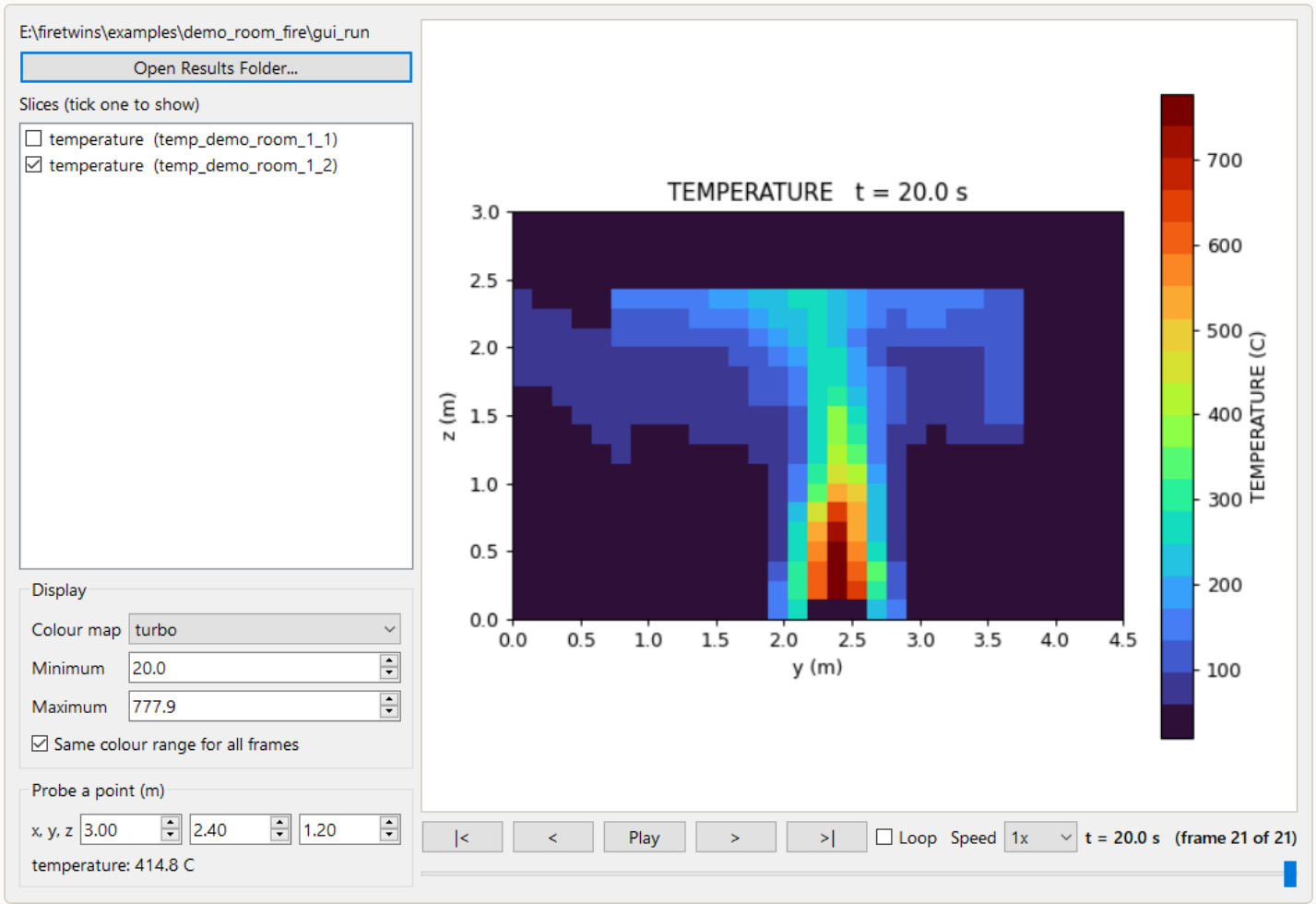
Start of the run: room at ambient temperature.



Mid-run, t = 10 s: the plume reaches the ceiling and a hot layer begins to form.



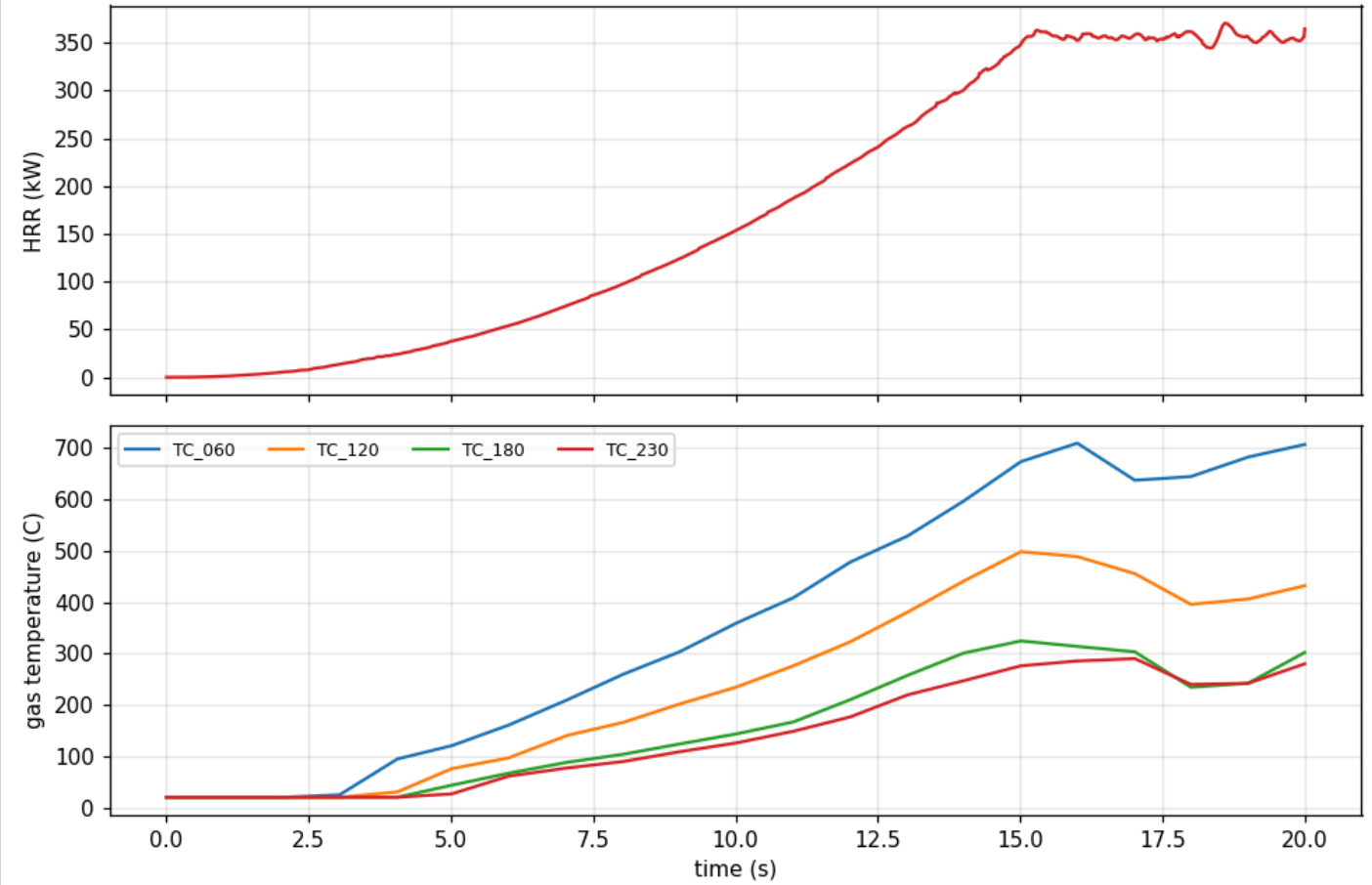
End of the run, t = 20 s: hot plume of about 780 °C at its base; the probe reads 415 °C at 1.2 m above the burner.



The second slice, on the plane x = 3.0 m, selected from the list on the left.

9b · Heat release and temperatures

The top curve is the heat release rate. It follows the t-squared growth and levels off near 360 kW once the burner is fully involved at 15 s. The lower curves are the four thermocouples. The hottest gas is just above the flame; temperature falls with height as the plume mixes with room air.



Heat release rate (top) and gas temperature at four heights (bottom), from the run's output files.

707 °C
thermocouple at 0.6 m, 20 s

432 °C
thermocouple at 1.2 m, 20 s

302 °C
thermocouple at 1.8 m, 20 s

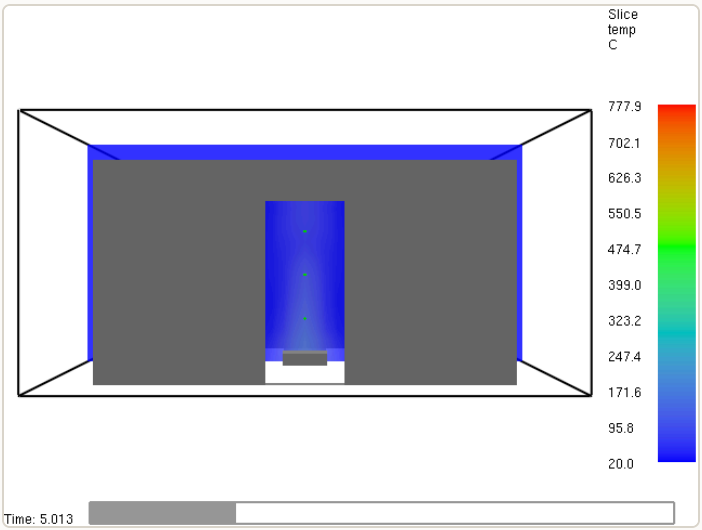
280 °C
thermocouple at 2.3 m, 20 s

Detector response

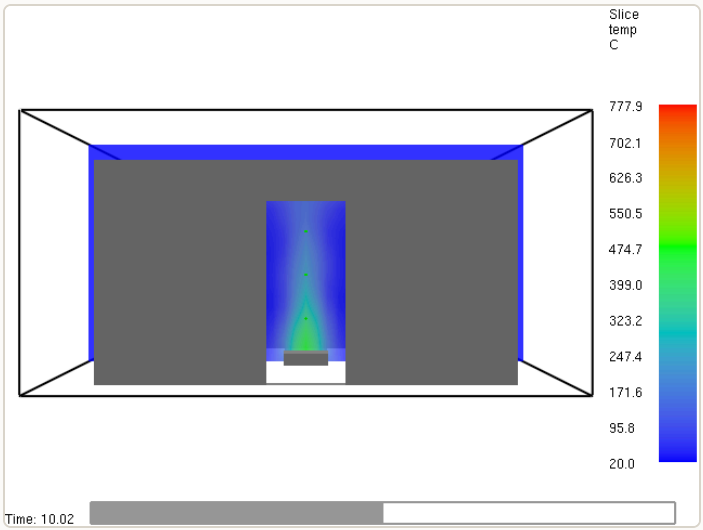
FDS reports the heat detector activating at 9.7 s, when the fire was about 145 kW. The recorded link temperature (saved once per second) passes 74 °C between 10 and 11 s.

9c · The same fire in Smokeview

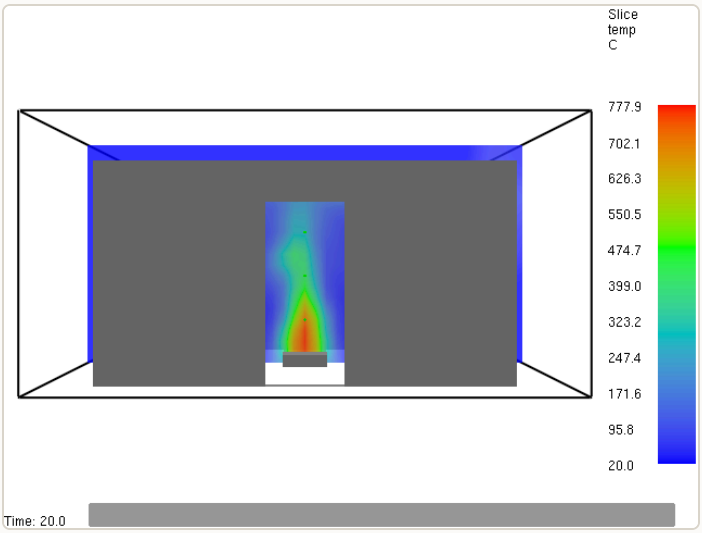
PyFDS can also drive FDS's own viewer, Smokeview, to make screenshots at chosen times. These show the temperature slice at 5, 10 and 20 s looking at the room from the south. The grey surfaces are the walls and the white gap is the door.



5 s



10 s



20 s

Values read from the results at 20 s: **415 °C** 1.2 m above the burner, **229 °C** under the ceiling above it, **45 °C** in the doorway.

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10 · The 3D Results Viewer

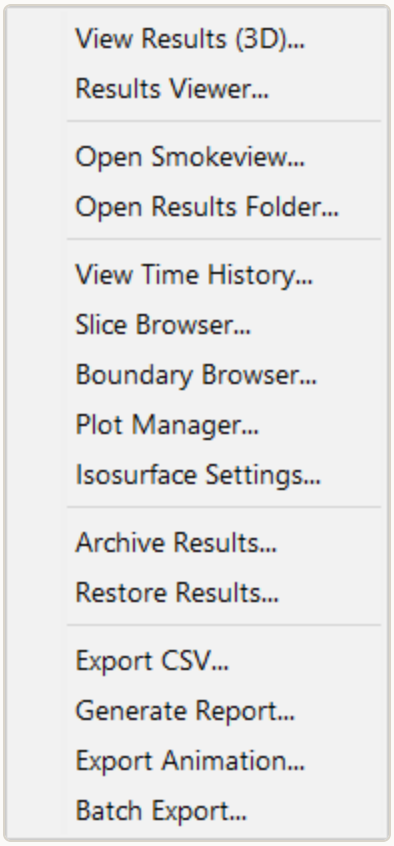
Section 9 looked at the run on flat slices. The **3D Results Viewer** is a second window, in the style of Smokeview's navigation view: the room, the fire and the data all sit in one 3D scene that you can

turn, cut open, play like a film and save as pictures or a video. Everything below uses the same run as section 8 (`gui_run/demo_room.smv`), and every screenshot is the real window.

10a · Open the viewer

`MENU` Results > View Results (3D)...

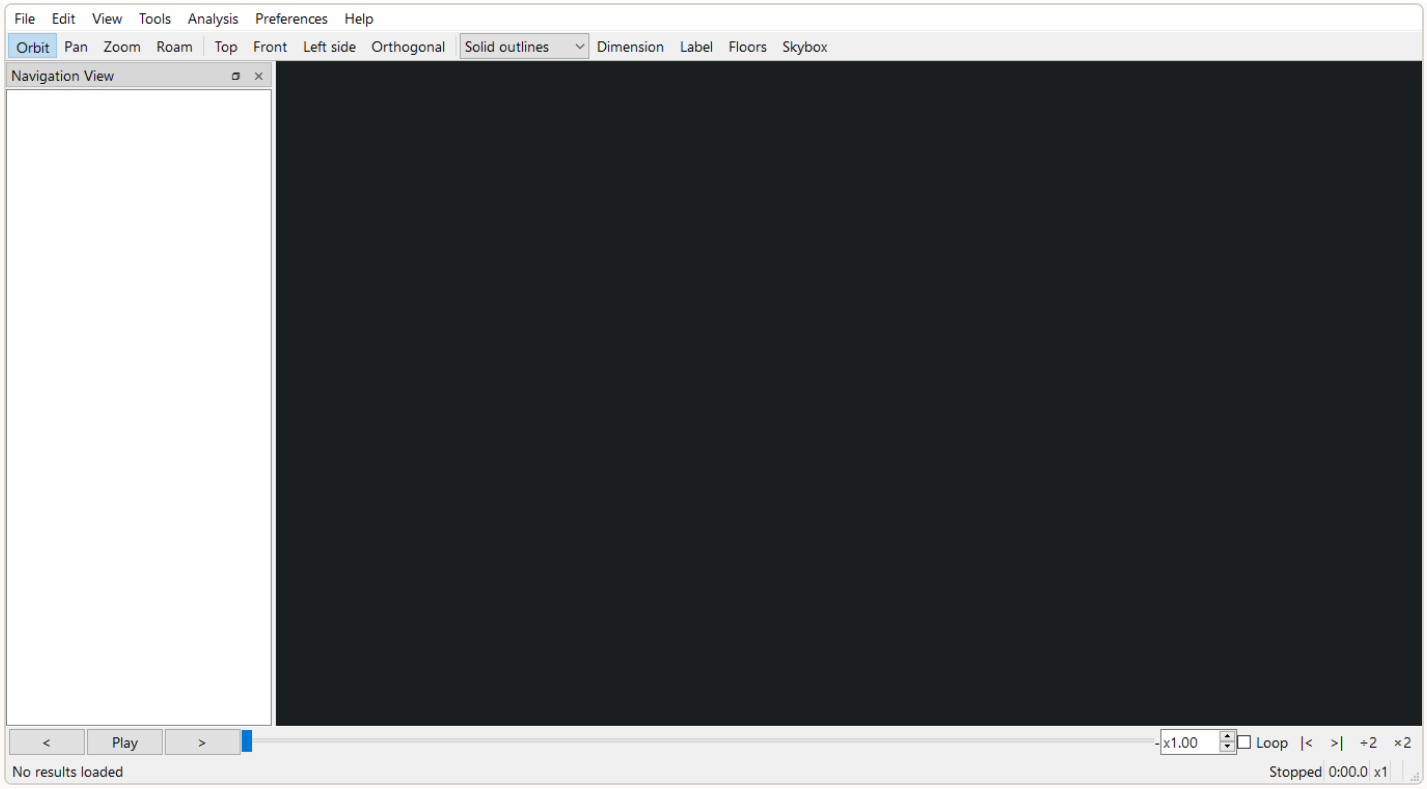
The entry sits at the top of the Results menu of the main window.



Results > View Results (3D)...



The viewer's own File menu: open or load results, load a CSV file, save screenshots and movies.

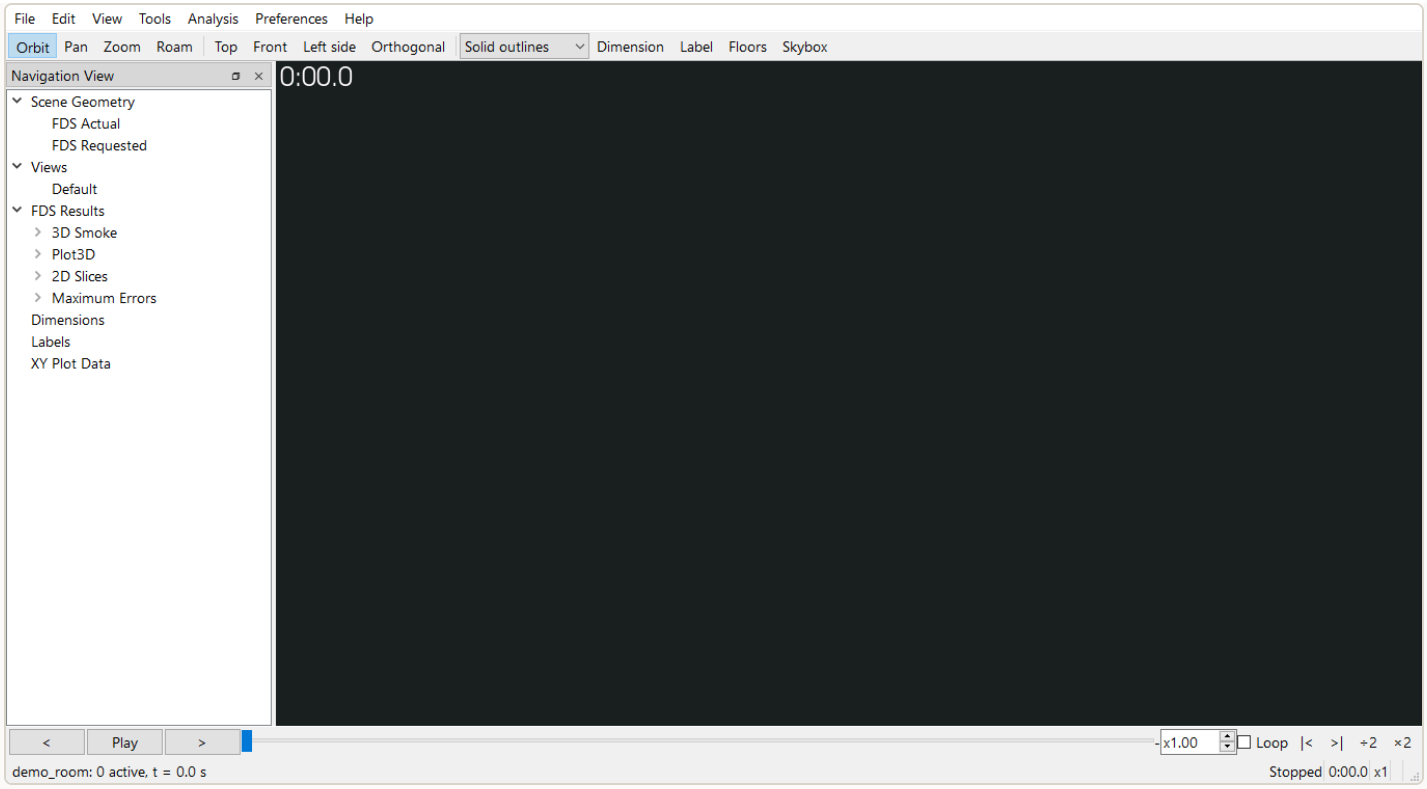


The viewer before anything is loaded. **Left:** the Navigation View (a tree of everything that can be shown). **Centre:** the 3D scene. **Bottom:** the time bar with Play, a slider, speed and Loop. **Top:** menus and a toolbar with the camera tools and rendering mode.

10b · Load the results of the run

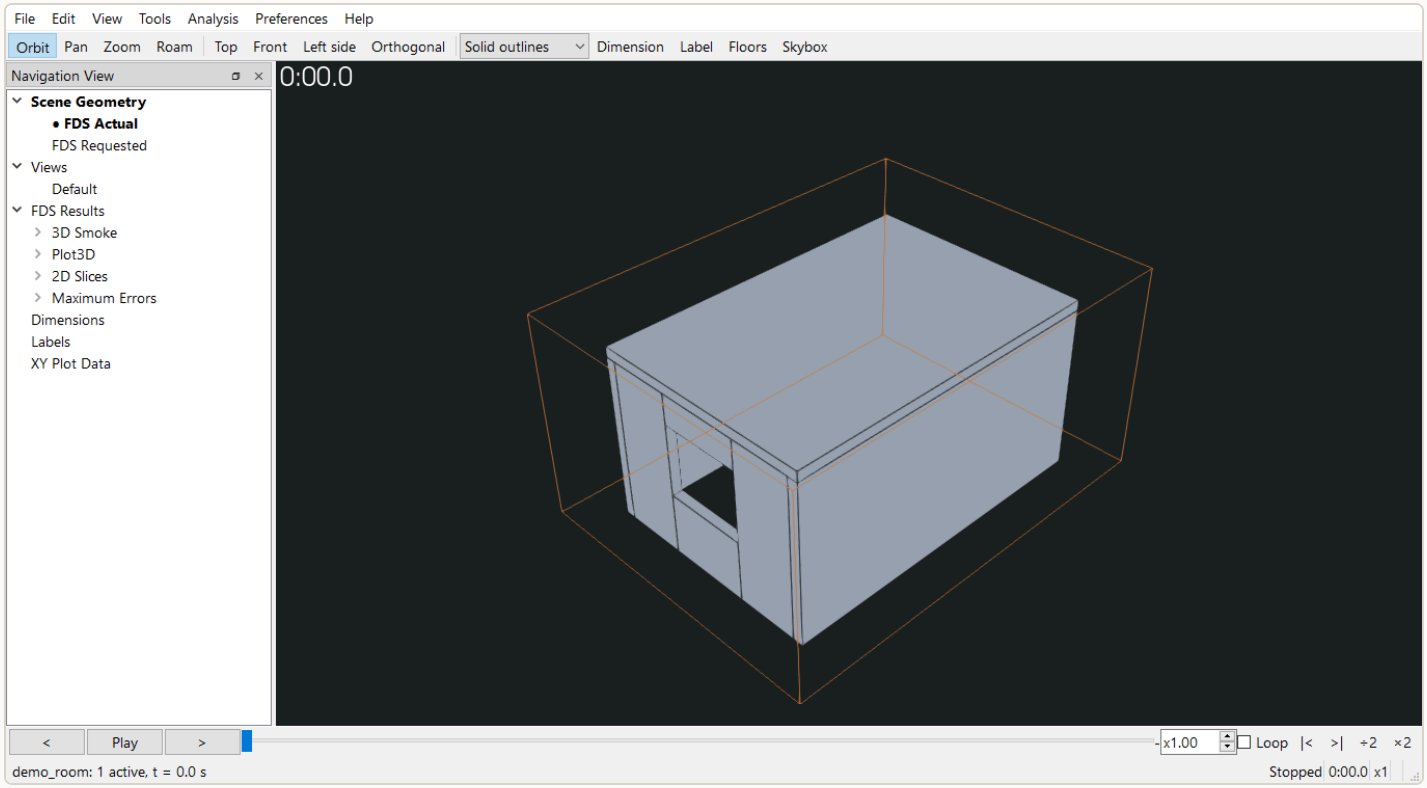
`MENU` File > Open Results (.smv)... then pick `demo_room.smv` in the run folder

The `.smv` file is the index FDS writes next to its data. The viewer reads it and builds the tree: scene geometry, **3D Smoke** (smoke and flame), **Plot3D** (the whole room at chosen times), **2D Slices** (the slices you asked for in step 5), and the solver's **Maximum Errors**. Nothing is drawn until you switch an item on. The case here has two slices, three kinds of smoke data and four Plot3D quantities.



Results loaded. The tree is full; the scene is empty until something is switched on.

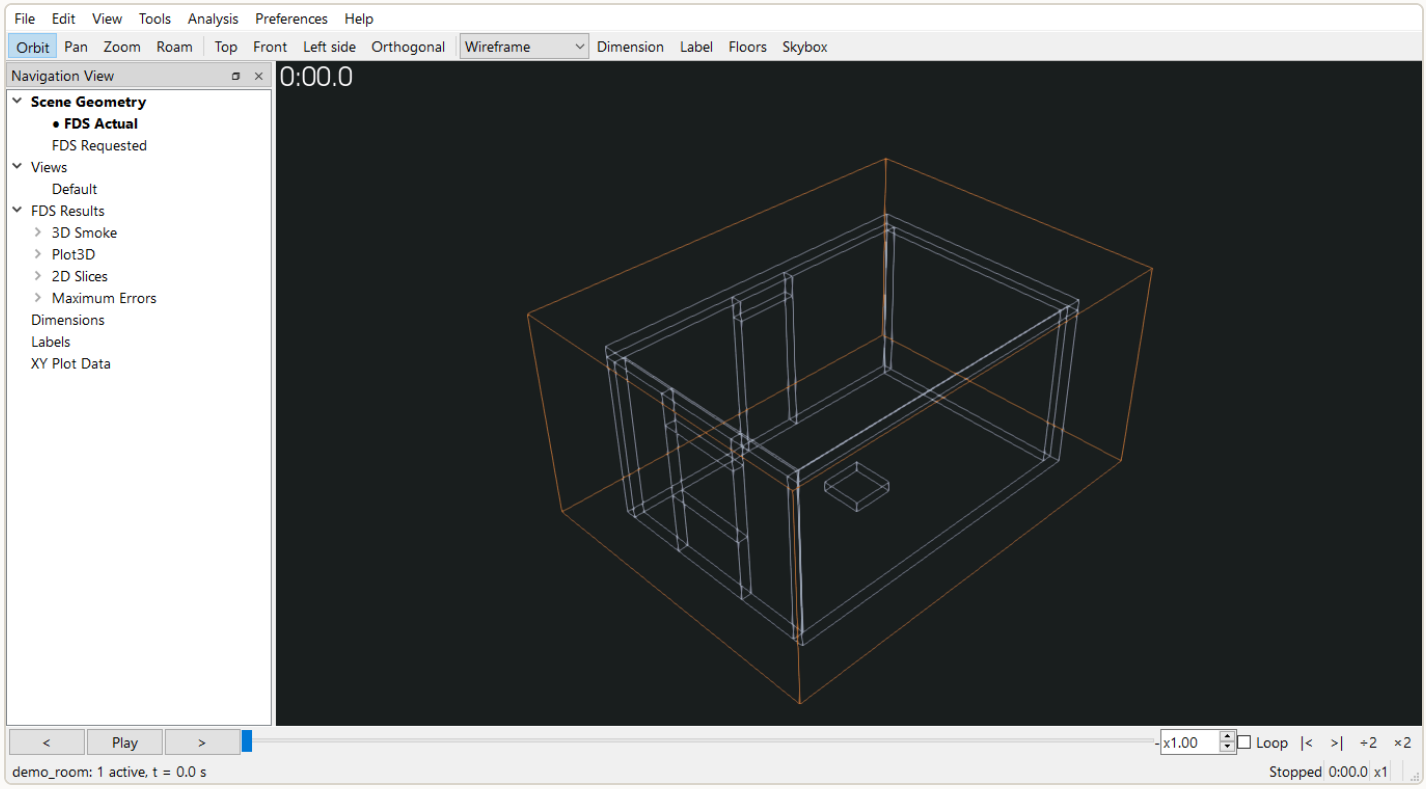
Double-click an item in the tree to switch it on or off (active items are shown in bold). Switch on **FDS Actual** under Scene Geometry to see what FDS really built: the walls, door, window and burner, inside the outline of the computational mesh.



FDS Actual geometry. The orange outline is the open boundary of the domain; the grey boxes are the walls, ceiling, door and window opening.

Rendering mode

The box in the toolbar chooses how solids are drawn (Solid, Solid outlines, Realistic, Realistic outlines, Wireframe). The next figures use **Wireframe** so the inside of the room stays visible.

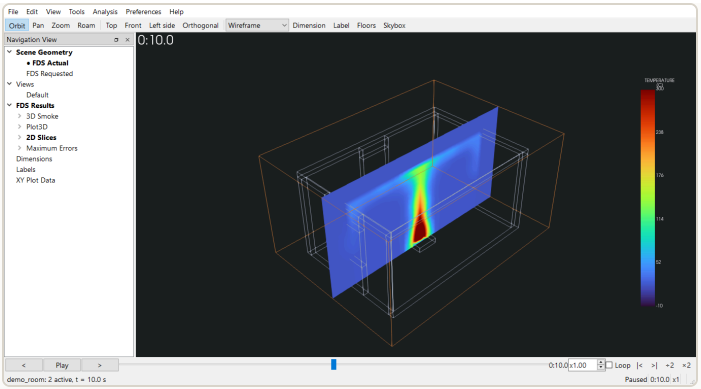


Wireframe: walls drawn as edges only.

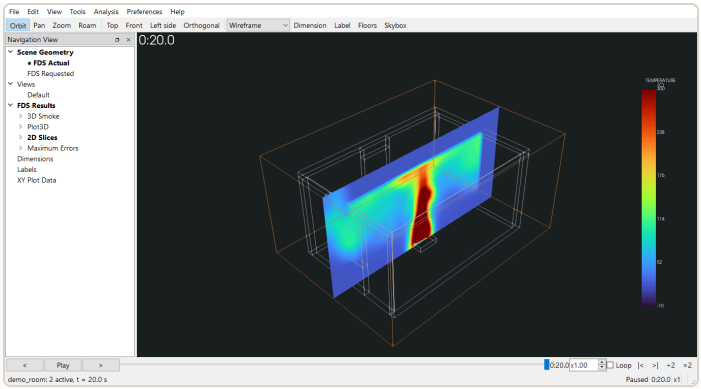
10c · Slices and the time bar

`TREE` 2D Slices > TEMPERATURE > SLICE_T_Y > Mesh 1 (double-click) · `TIME BAR` slider, Play, speed, Loop

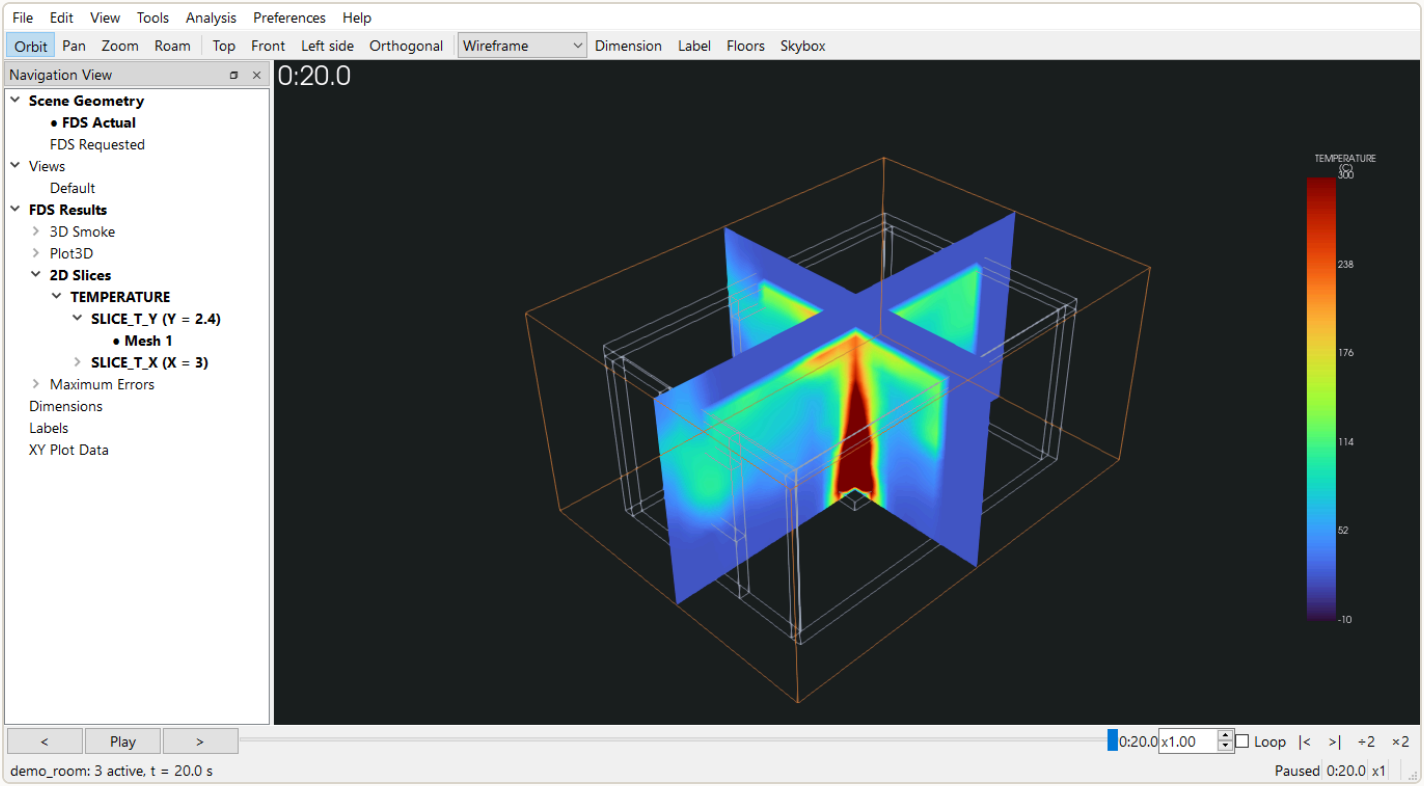
A slice is a flat cut through the room that shows one quantity in colour. Switching it on adds a **colour bar** at the right and a clock at the top left. Drag the slider to any time, step with **<** and **>**, or press **Play**. The speed box (×1.00) and the ÷2 and ×2 buttons change the playback speed; **Loop** repeats.



t = 10 s: the plume rises from the burner and starts to spread under the ceiling.



t = 20 s: a hot layer under the whole ceiling.

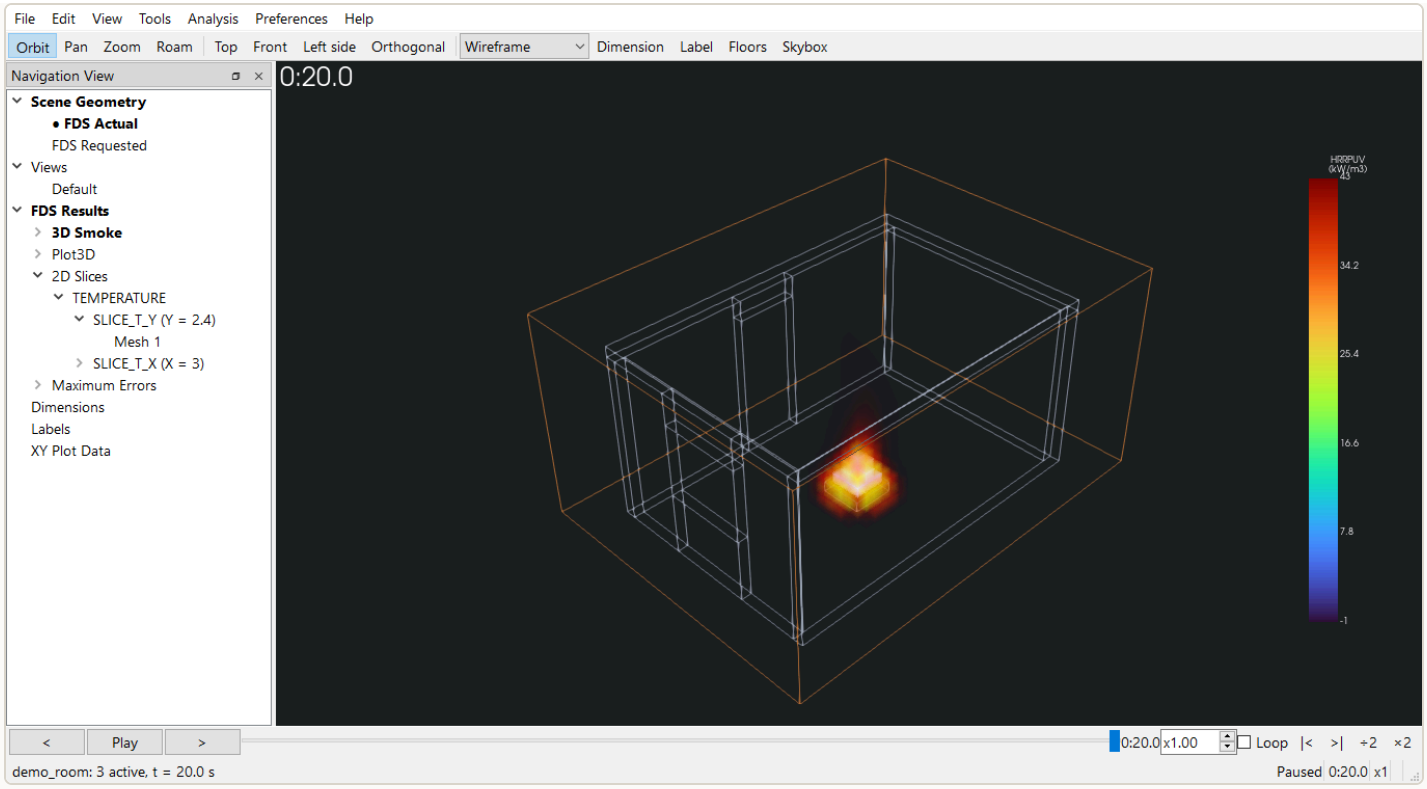


Both slices on at once (planes y = 2.4 m and x = 3.0 m). They cross exactly in the plume.

10d · 3D smoke and flame

TREE 3D Smoke > HRRPUV > Mesh 1 (flame) and SOOT DENSITY > Mesh 1 (smoke)

3D Smoke shows the fire itself as a volume, not a flat cut. **HRRPUV** (heat release rate per unit volume) is the flame: the brighter, the more heat is released in that spot. **SOOT DENSITY** is the smoke. In this short run with a clean-burning gas fire there is almost no soot, so the flame is the part to look at. Look at it from different sides by dragging with the mouse (Orbit), or use the Top, Front and Left side buttons.

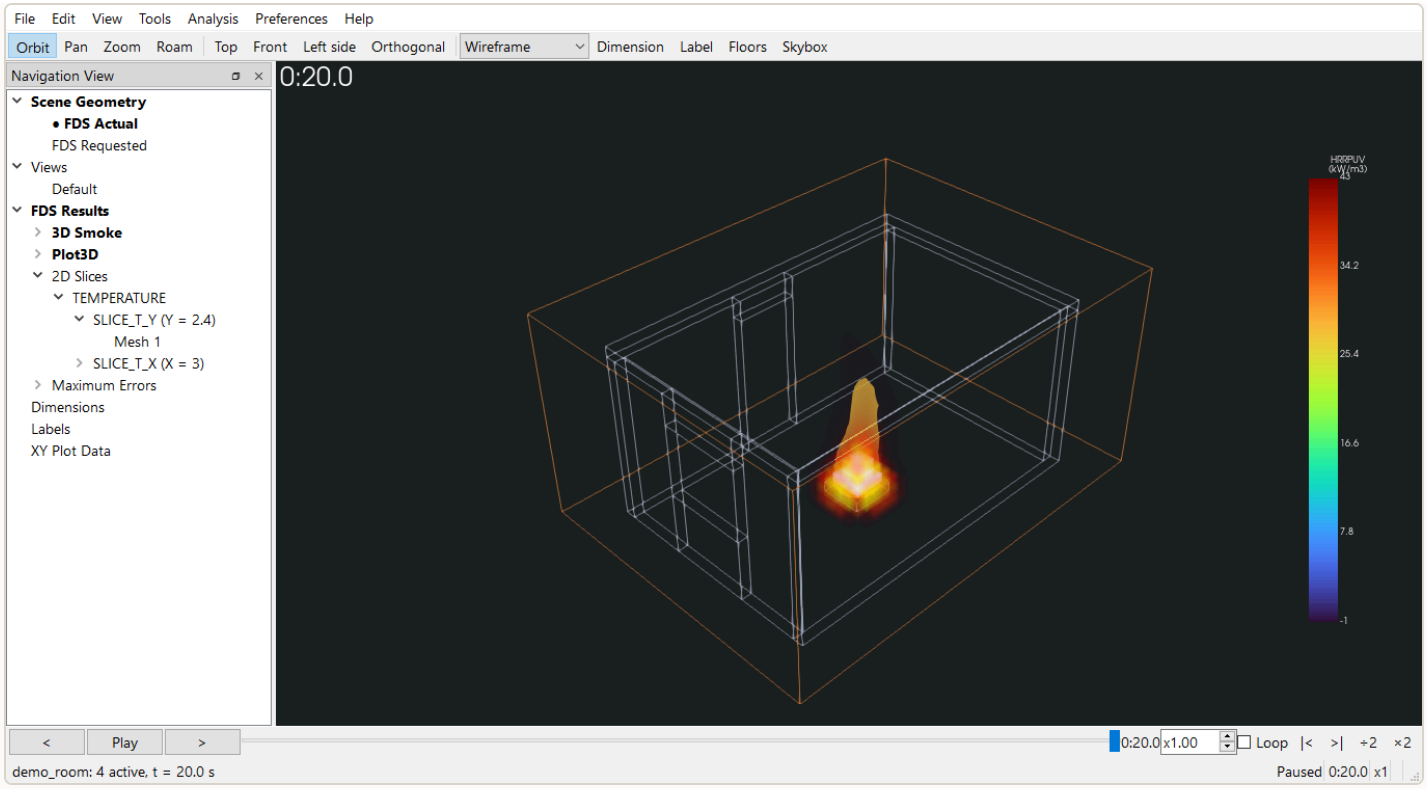


The flame at t = 20 s, drawn as a volume above the burner.

10e · Isosurfaces from Plot3D

MENU Tools > Data3D: Add Isosurface… · choose the quantity (Plot3D > TEMPERATURE), enter the value (300), then switch the new item on in the tree

Plot3D files hold the whole room at a few times. An **isosurface** is the skin of all points that have exactly one value, here 300 °C. It shows the shape of the hot plume as a solid body.



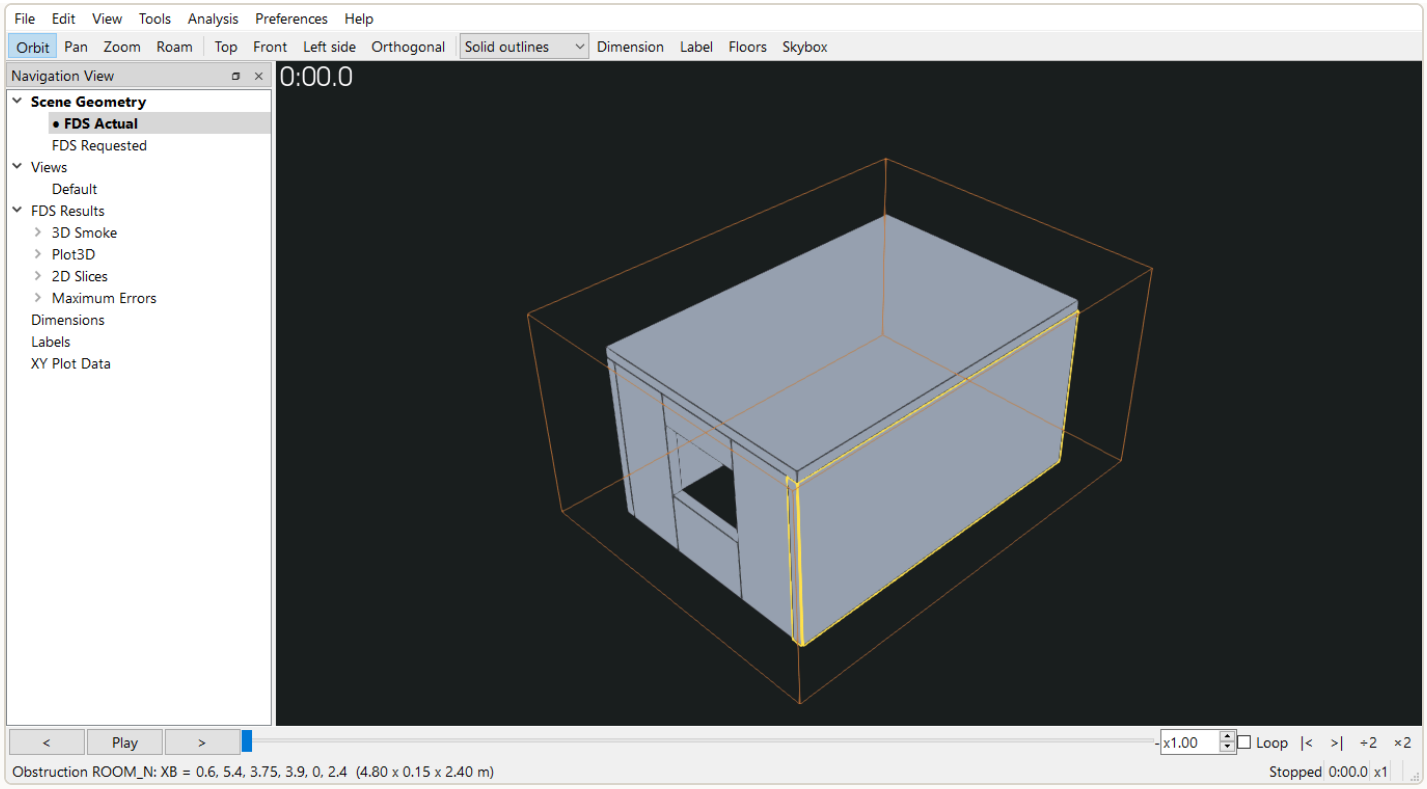
The 300 °C surface (yellow) at t = 20 s: a tall column over the burner that has not yet reached the ceiling.

10f · Select objects by clicking

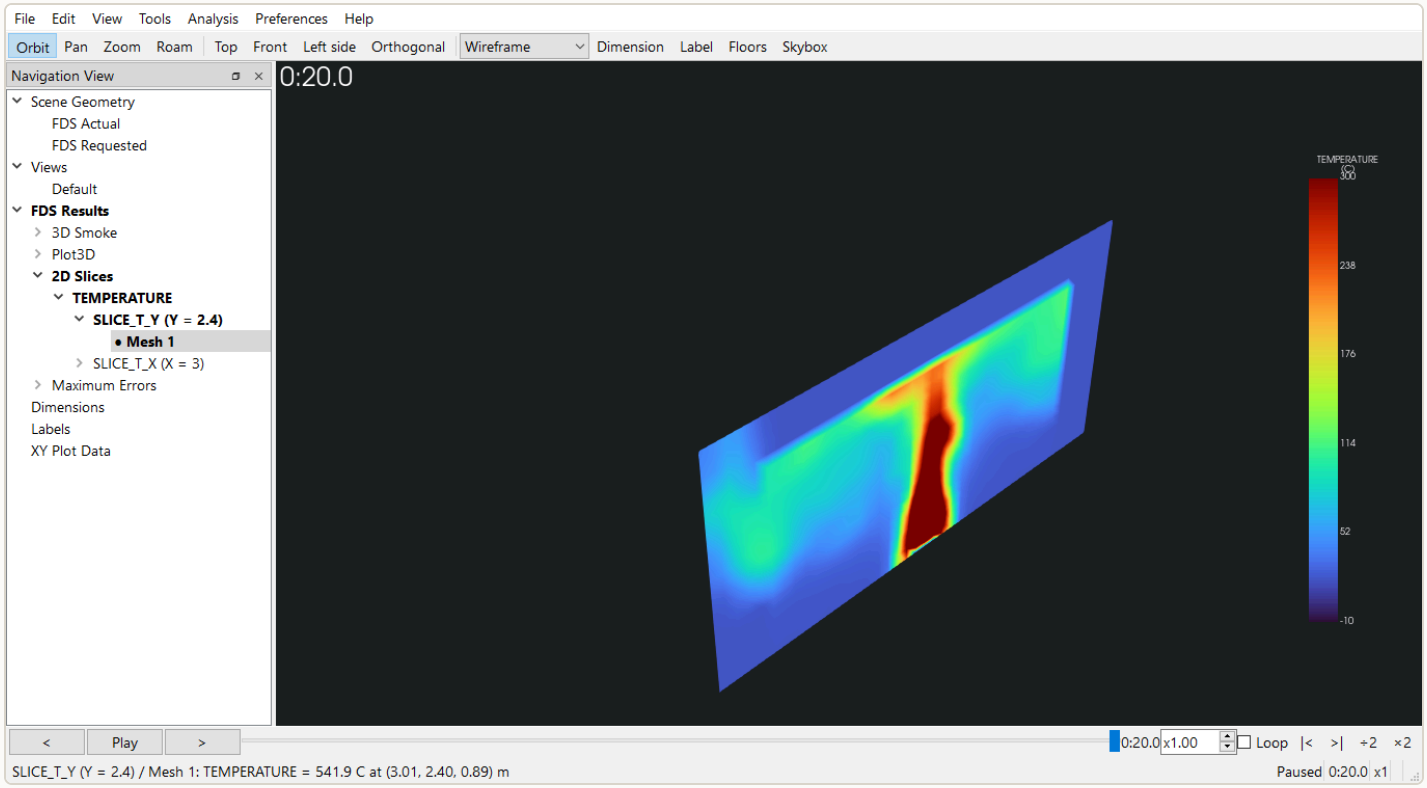
`MOUSE` click an object in the 3D scene · `ESC` or click empty space to clear

Click any object in the scene to find out what it is. A click is a press and release without moving the mouse; if you drag, the view turns instead. The object you hit is **outlined in yellow**, its row in the Navigation View is selected, and the **status bar** at the bottom describes it:

- **A wall, vent or other solid:** its name, its position (XB) and its size. In the figure below, the click on the north wall reports `Obstruction ROOM_N: XB = 0.6, 5.4, 3.75, 3.9, 0, 2.4 (4.80 × 0.15 × 2.40 m)`.
- **A slice or 3D data:** the value under the pointer. Clicking the temperature slice at the base of the plume reports `TEMPERATURE = 541.9 C at (3.01, 2.40, 0.89) m`.
- **An isosurface:** its level.



A click on the north wall: yellow outline, the tree row is selected, and the status bar gives the name and size.



A click on the slice at 0.89 m above the floor: the status bar reads 541.9 °C at that point (t = 20 s).

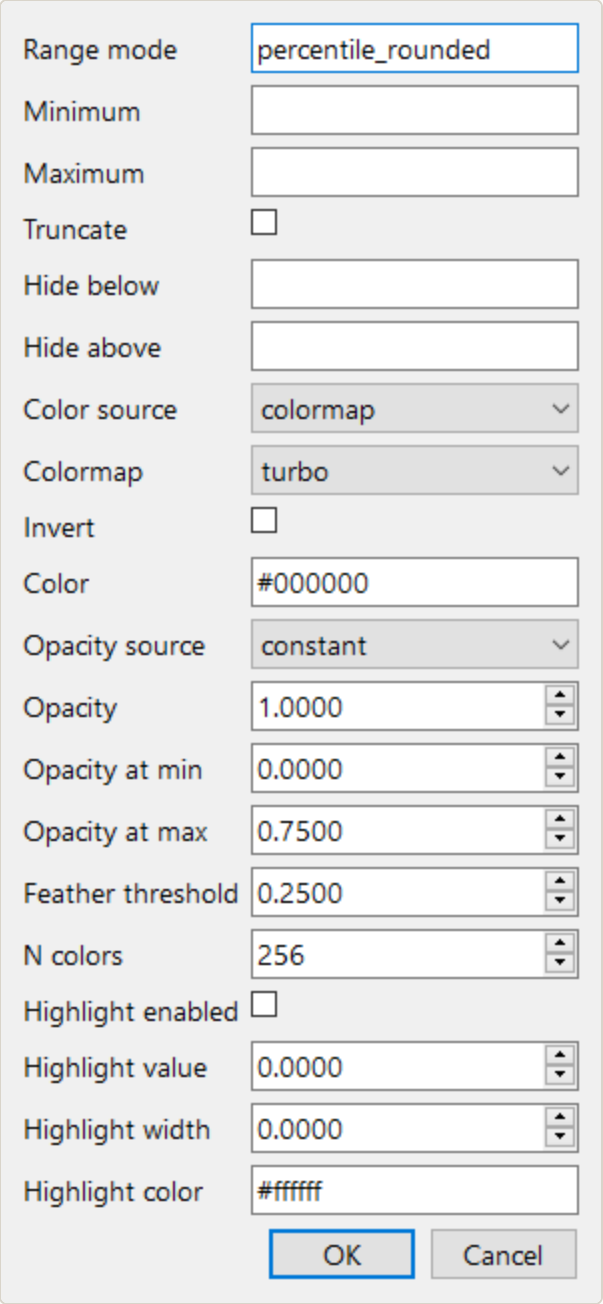
Good to know

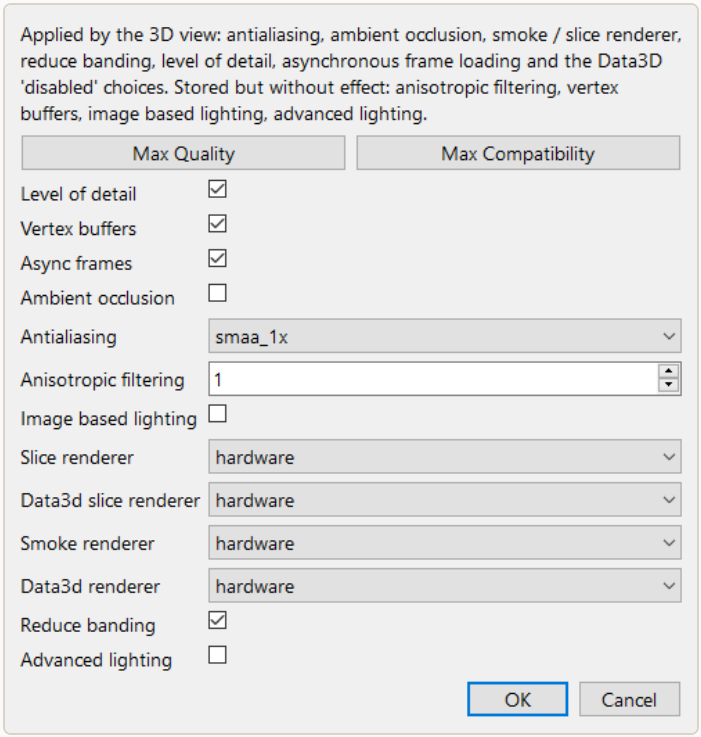
The open sides of the domain (the orange outline) are not clickable, so they never block the objects behind them. A click selects one object at a time; the 3D flame and smoke volumes cannot be selected by clicking.

10g · Colour bar and rendering quality

MENU Preferences > Colorbar... (pick the quantity first) · Preferences > Rendering...

The **Colorbar** dialog sets the range of the colours (minimum, maximum, or automatic), the colour map (turbo, viridis, jet, ...), whether values below or above a limit are hidden, and how see-through the data is. The **Rendering** dialog controls the picture quality. **Max Quality** switches on everything the graphics card can do and **Max Compatibility** picks the safe options for an older machine; the note at the top says which options are applied.





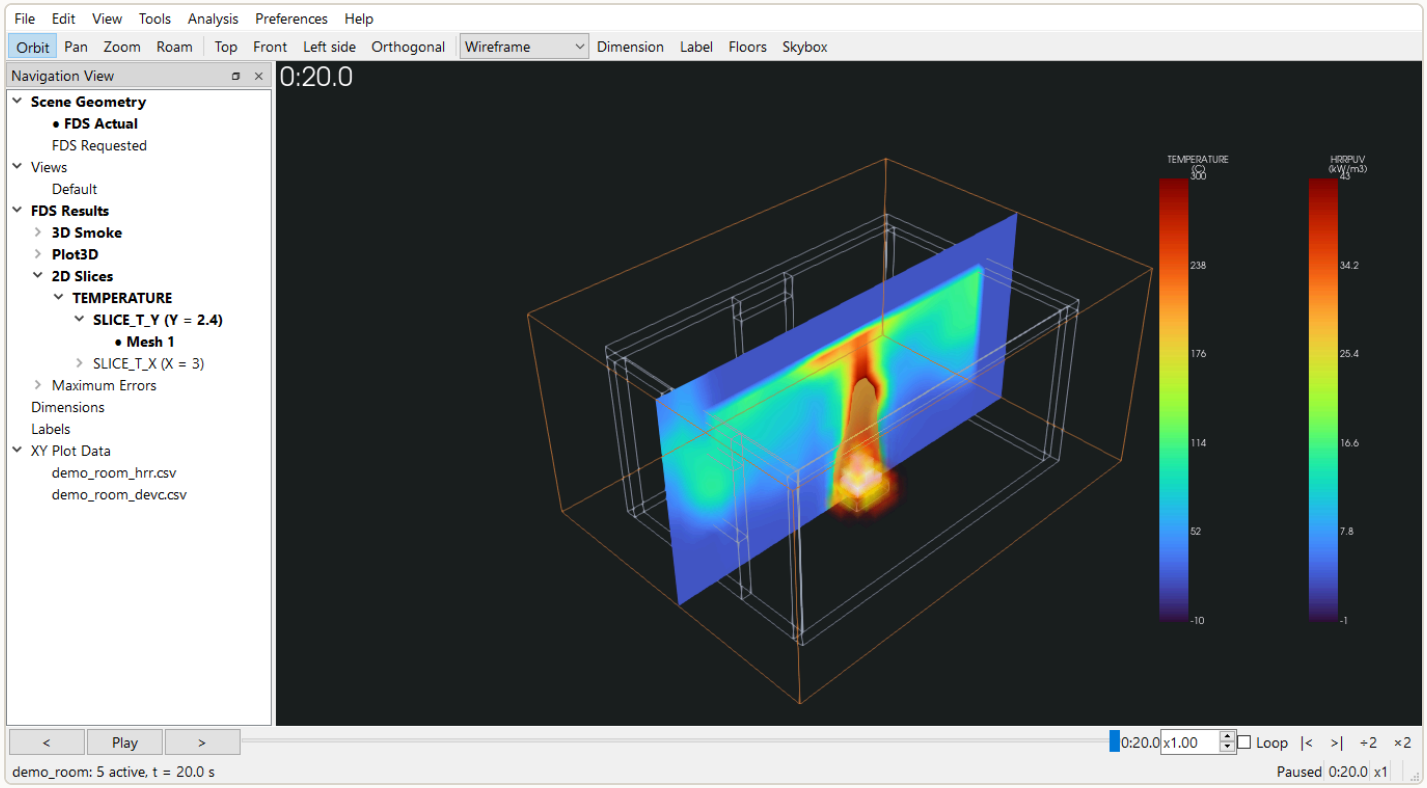
Rendering preferences with the two quality presets.

Colorbar settings for TEMPERATURE.

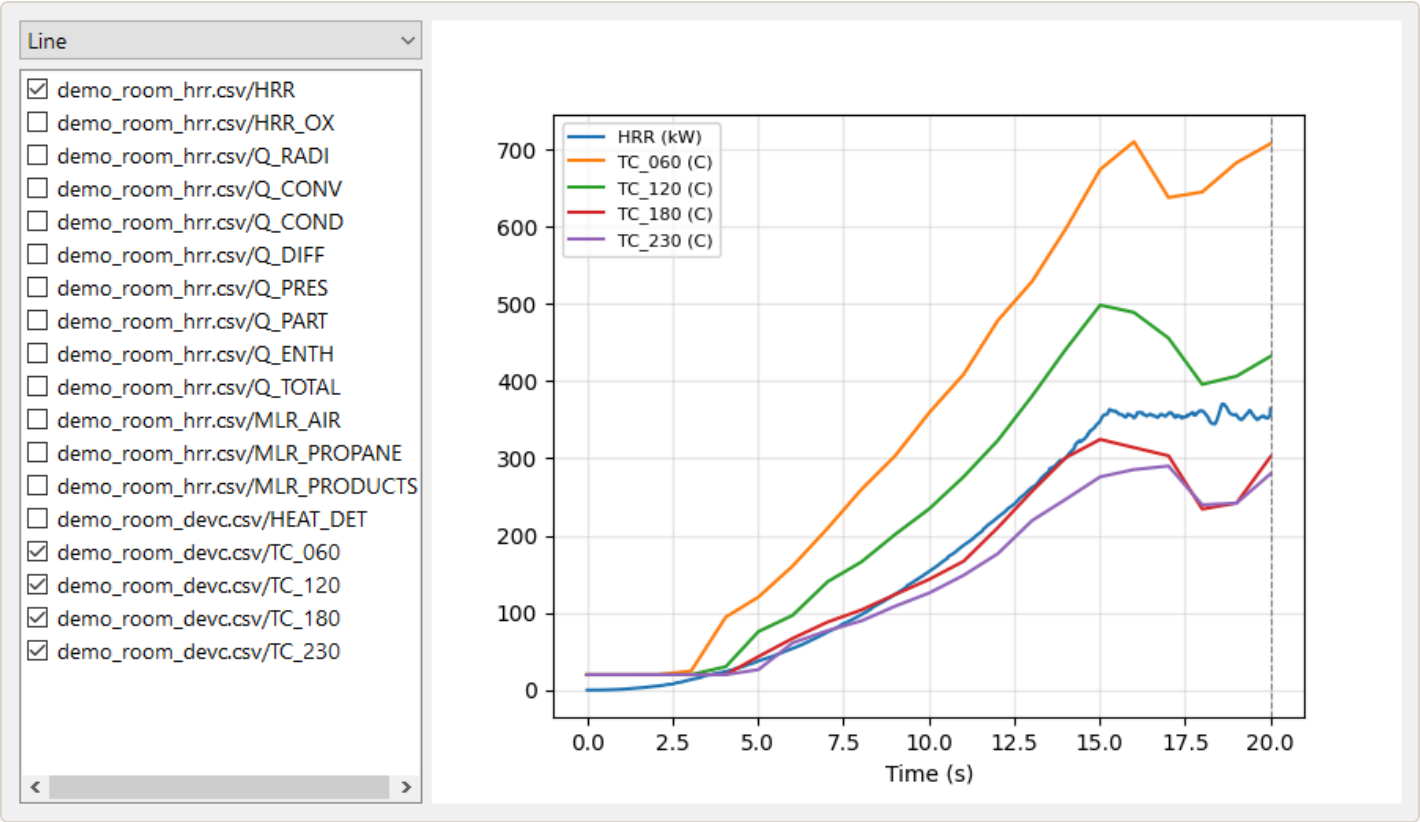
10h · Plots from the output files

`MENU` File > Load CSV File... (twice: `demo_room_hrr.csv`, `demo_room_devc.csv`) · Tools > Time History Plot...

The heat release file and the device file written by FDS can be loaded next to the 3D data. They appear under **XY Plot Data**. The plot window lists every column; tick the ones you want. A dashed line marks the time that the 3D scene is showing, so the curves and the picture can be read together. The plot type can be Line, Stem or Step, and **F11** saves the plot as an image.



Both CSV files loaded and listed in the tree.

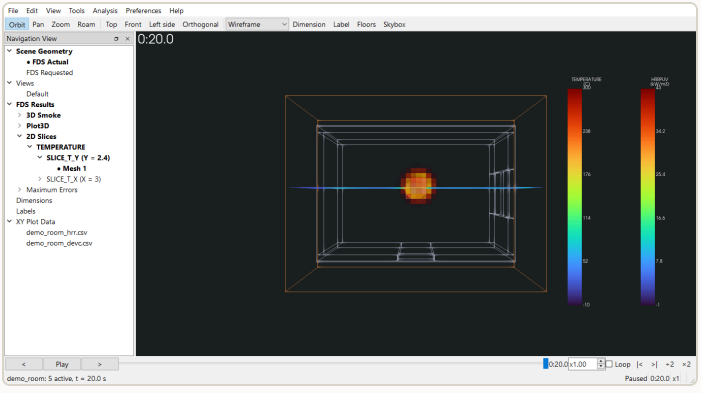


The heat release rate (blue, kW) and four thermocouple temperatures (°C). The same curves as in 9b, now inside the viewer.

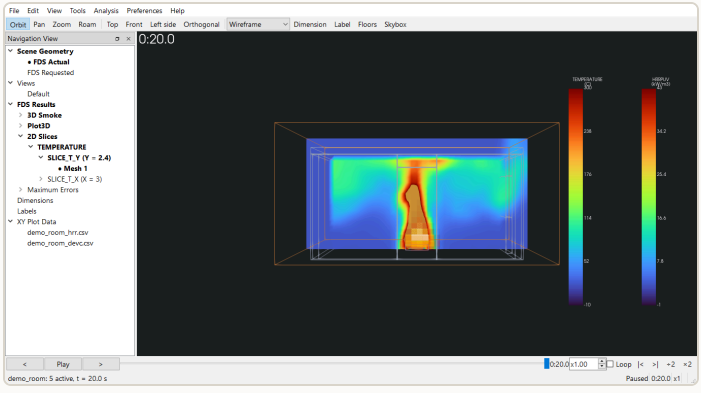
10i · Views, section planes, dimensions and labels

MENU View > Top / Front / Side / Reset Camera · View > Set Section Plane... · Tools > Add Dimension... / Add Label... · View > New View...

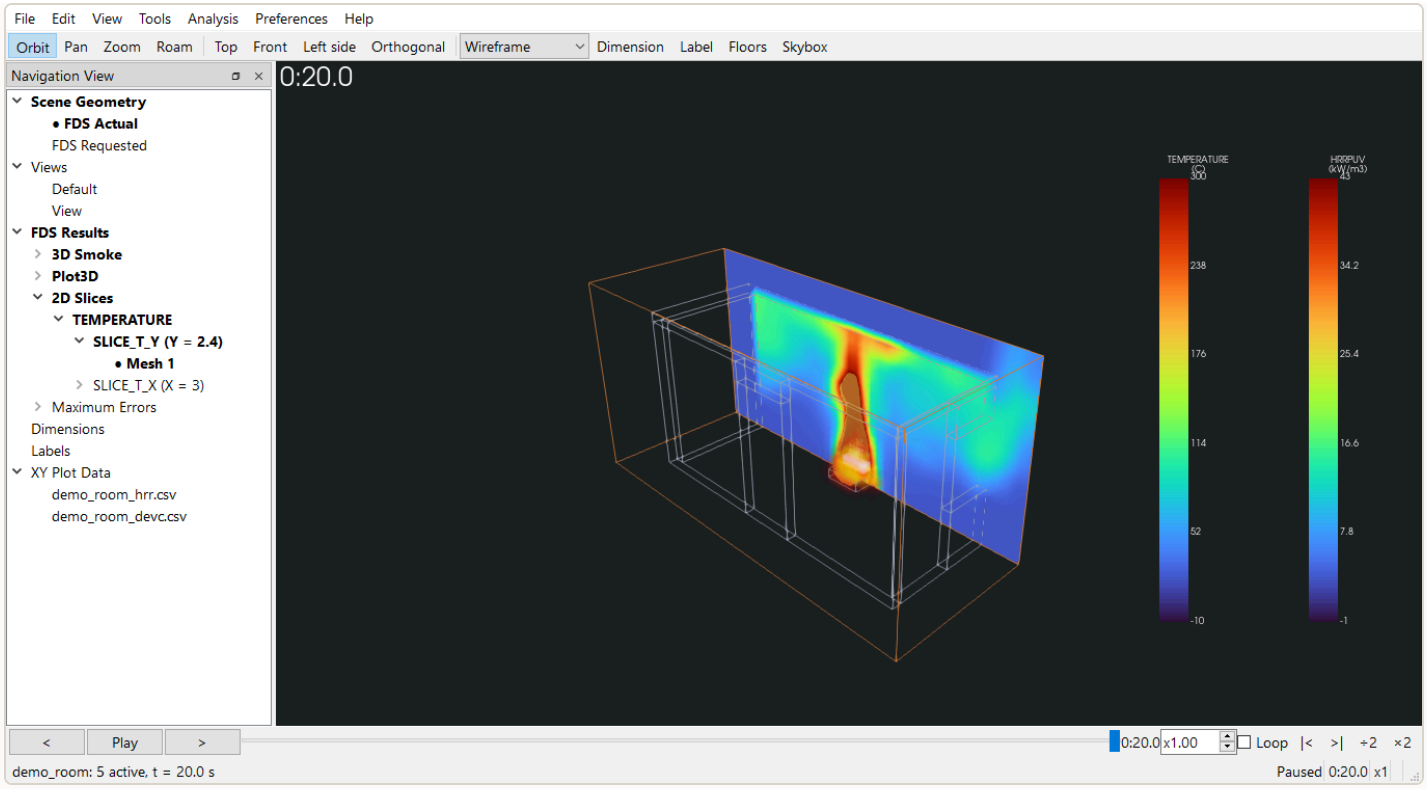
Use the toolbar or the View menu to jump to the standard directions. A **section plane** cuts the model away on one side, like slicing a cake, so you can see inside without moving the camera. A **dimension** draws a measured line between two points; a **label** pins a text to a point. **New View** stores the current camera position under a name, so you can come back to it, or use it for screenshots and movies.



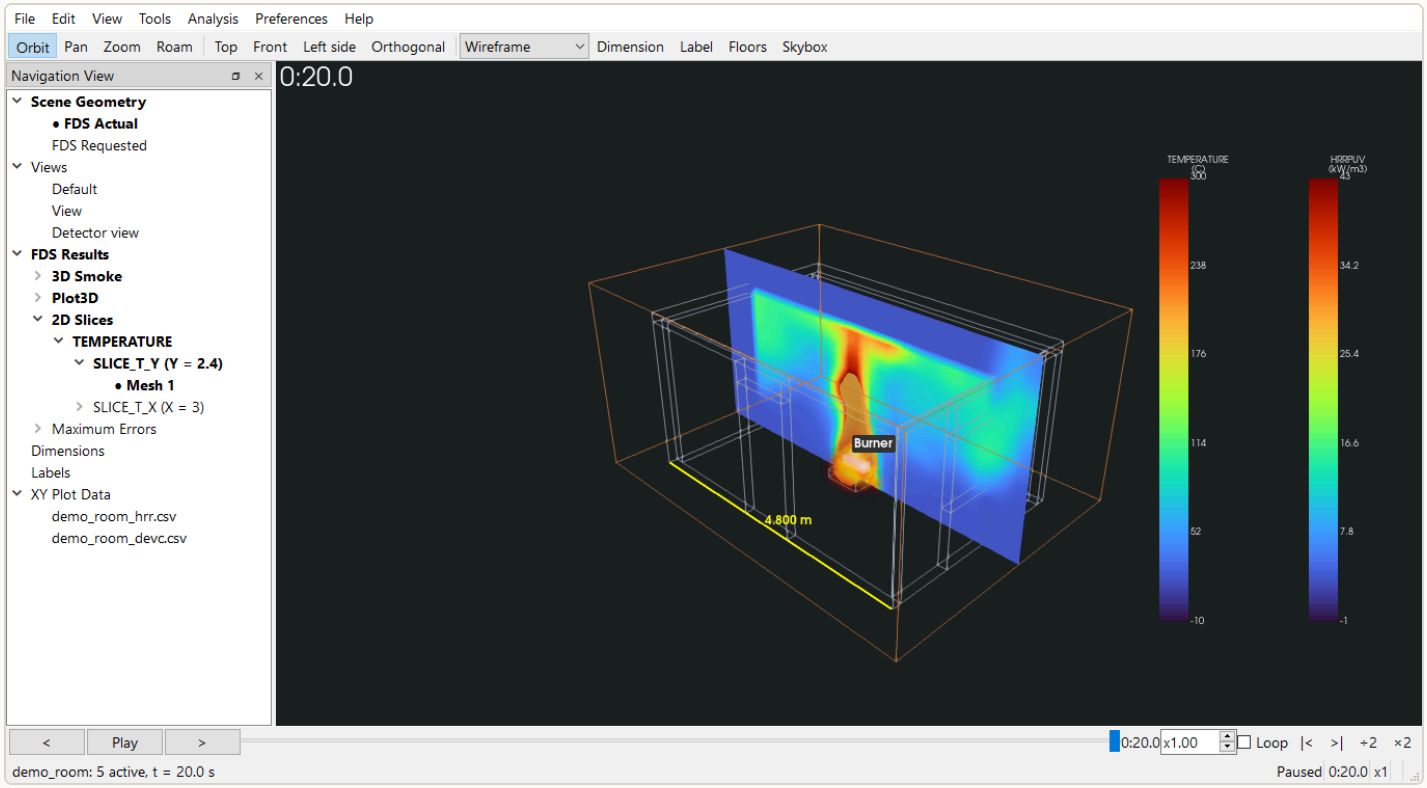
Top view.



Front view.



A section plane at y = 2.4 m: the far half of the room is removed and the plume is seen from the side.



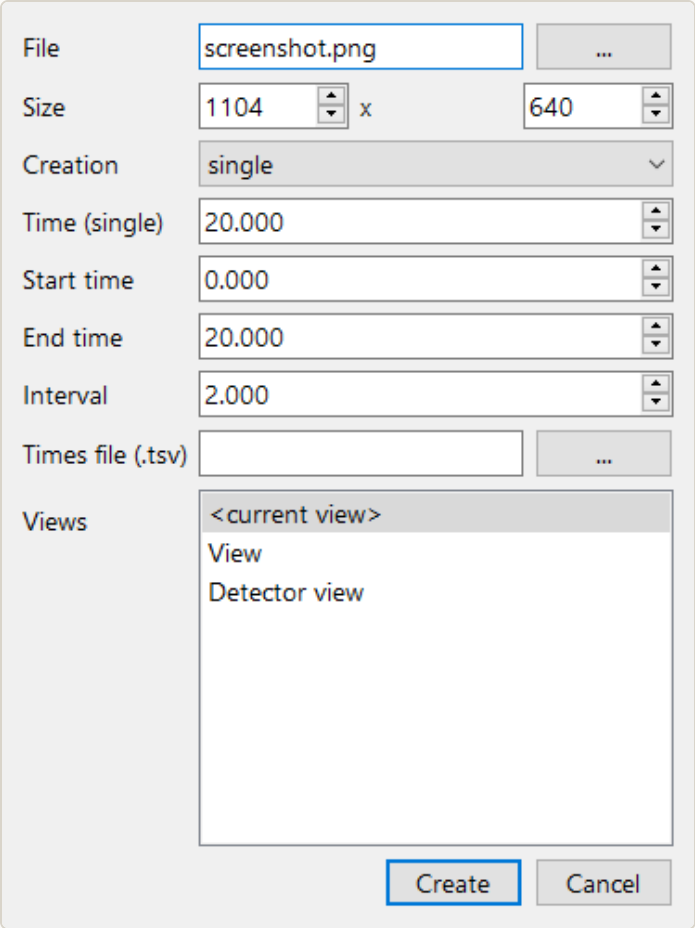
A dimension (4.800 m along the south wall) and a saved view named “Detector view”, both listed in the tree.

10j · Screenshots and the animation

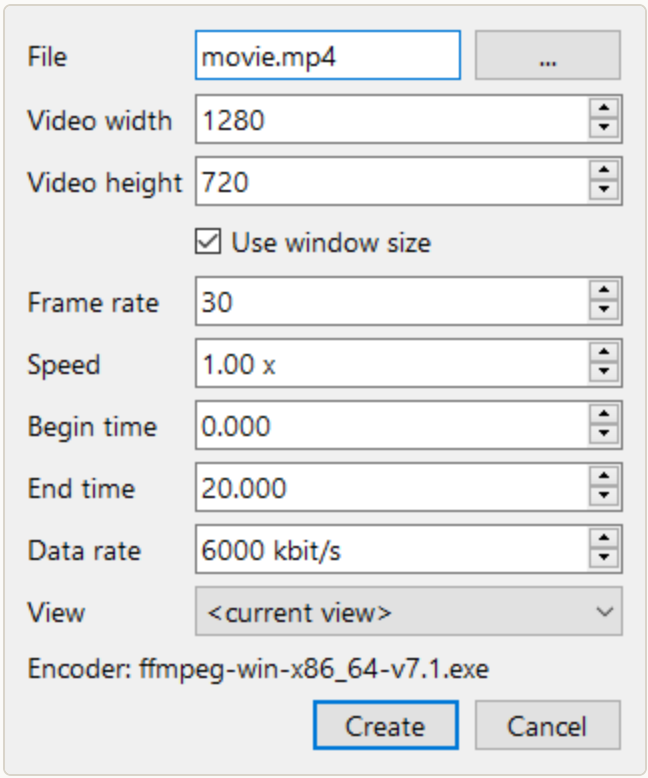
MENU File > Save Screenshot... (Ctrl+P) · File > Save Movie... · File > Save 360 Movie...

Screenshot. The dialog asks for a file name, the picture size, and how many pictures to make: a **single** picture at one time, **multiple** pictures between a start and end time at a fixed interval, or pictures at the times listed in a text file. You can also choose which saved views to photograph.

Movie. The dialog asks for the file name, size, frame rate, speed (2× means two seconds of fire per second of film), begin and end time, data rate and which view to use. The viewer draws each frame itself and joins them into an MP4 file. For this case the run was made into a film of **101 frames** at 10 frames per second and 2× speed (about 10 seconds long, 960 × 540 pixels). It is saved as `gui_viewer/demo_room.mp4` . **Save 360 Movie** does the same for a 360° picture that can be looked around in a VR or 360° player.

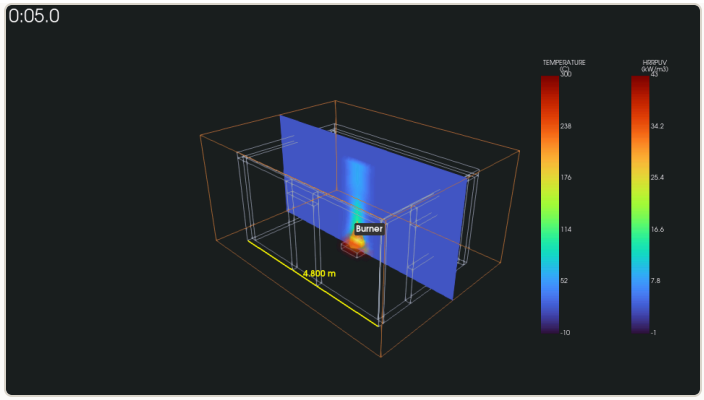


Save Screenshot dialog.

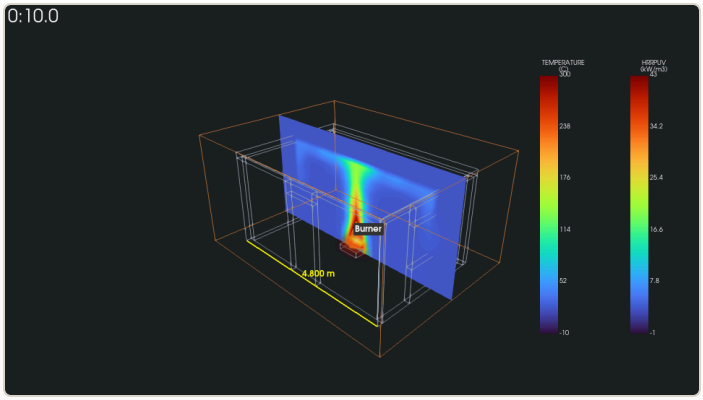


Save Movie dialog. The line at the bottom shows which video encoder was found.

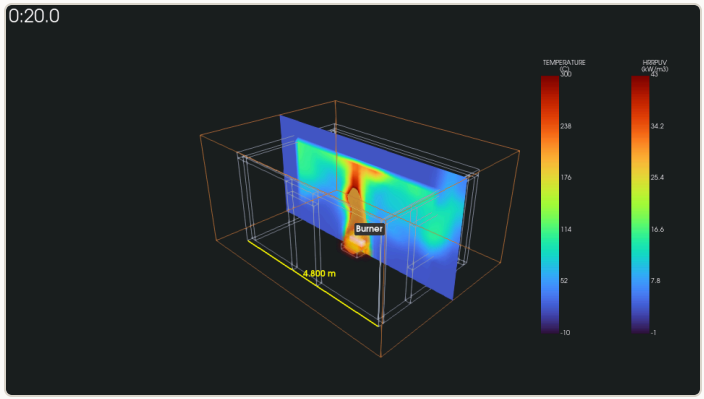
Three frames from the film (the temperature slice on the plane y = 2.4 m together with the flame, the burner label and the room dimension):



5 s



10 s



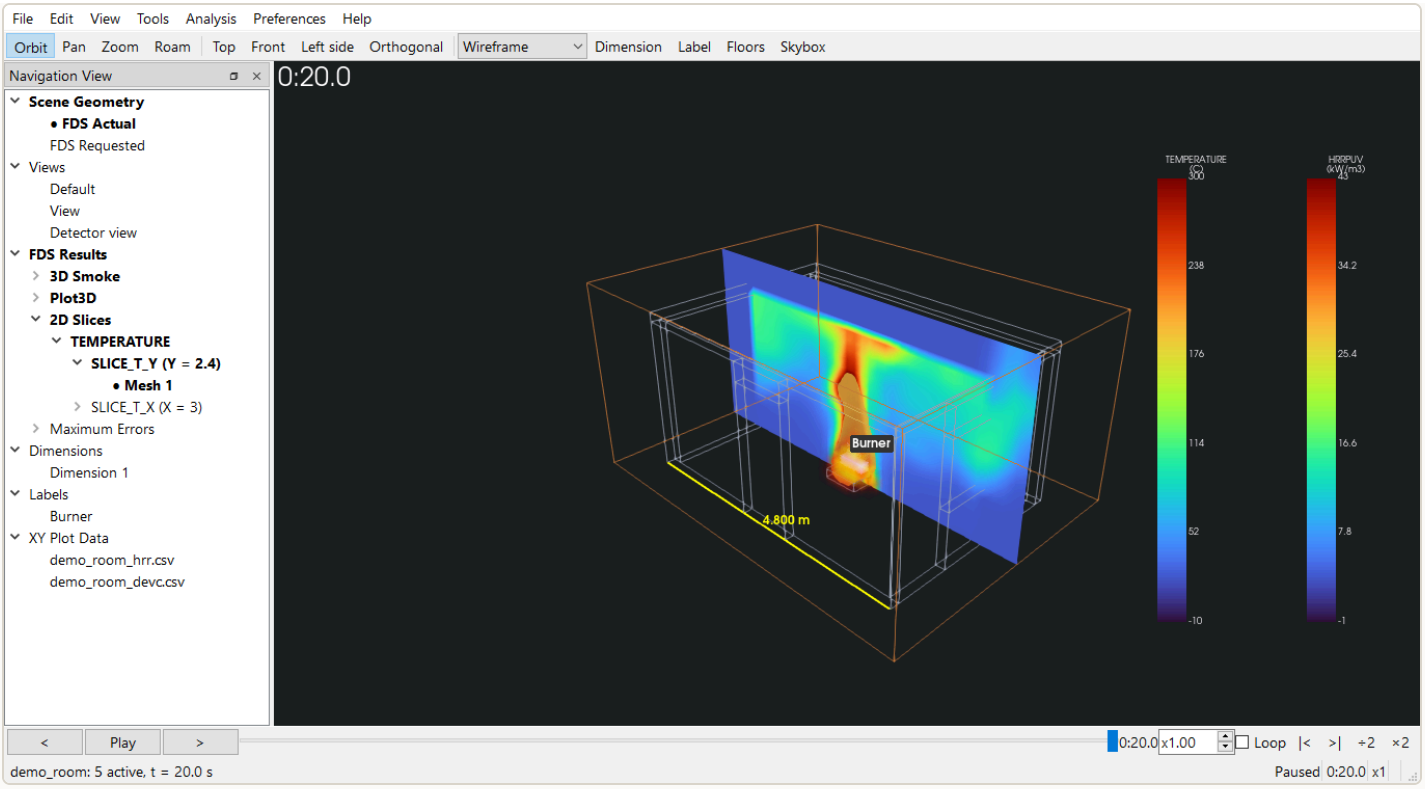
20 s

The same can be done without opening the window, for example for a batch of cases: `python -m pyfds.gui.results -movie -mvname film.mp4 demo_room.smv` (add `-mv360` for a 360° film, or `-screenshot` for pictures).

10k · Save the visualization

`MENU` File > Save As... → `demo_room.pyfdsv`

Everything set up in the viewer (which items are on, the colour bars, views, dimensions, labels, section planes, loaded CSV files, preferences) can be saved in one small text file with the extension **.pyfdsv**. Opening it later with File > Open Results Visualization brings back the same picture. The file refers to the run's folder; it does not copy the data.



Saved. The window title shows the file name; an asterisk appears there when there are unsaved changes.

Files, repeating, limits

What you receive

FILE	WHAT IT IS
`demo_room_gui.pyfds`	The project saved from the application. Open it to change anything and run again.
`gui_run/`	All FDS output: log, heat release file, device file, slice files, smoke data, and the FDS input that was run (`demo_room.fds` , plain text).
`gui/`	The screenshots in this manual, one per step.
`gui_results/`	Chart, Smokeview screenshots and a summary of the key numbers.
`gui_viewer/`	Screenshots of the 3D Results Viewer (section 10), three film frames, and `demo_room.mp4` , the animation.
`demo_room.pyfdsv`	The saved 3D visualization (section 10k).
`gui_viewer_walkthrough.py`	The script that operated the 3D viewer to make section 10; run it after `gui_walkthrough.py` .

`gui_walkthrough.py`	The script that operated the application window to produce this manual, so the whole case can be repeated.

Repeating the case

Open `demo_room_gui.pyfds` with File > Open, change a value (a larger fire, a wider door, a longer end time) and run again. To redo the whole walkthrough, including the screenshots, run `python examples/demo_room_fire/gui_walkthrough.py` (about four minutes, the window opens and operates itself).

Limits of this demonstration

- **It is a demonstration, not an engineering study.** Twenty seconds of a small fire in coarse cells shows the workflow. The fire is only borderline resolved (warning 1) and the room has not reached a steady state.
- **The 3D viewer is checked on this machine only.** Its screenshots were taken from the real window. Clicking an object in the 3D scene does not select it yet (use the tree), the animation is made frame by frame and not recorded live, and virtual-reality display is not included.
- **Library values are typical, not certified.** Material and detector entries are starting points to be replaced with product data.
- **Some dialogs show plain parameter names** (for example “Ijk”, “Xb”, “Tau q”) with a short explanation under the field. The meaning of each is given in the text above.
- **Movies need an encoder.** The MP4 in this manual was made with the ffmpeg encoder bundled with PyFDS. If no encoder is found, the viewer keeps the numbered picture files instead and says so.
- **Tested with:** FDS 6.10.1 and Smokeview 6 on Windows.

PyFDS demo case “demo_room”. Screenshots taken from the running application on 11 and 12 October 2026; numbers from the FDS output of that run.